
WHEN IS CUMULATIVE DOSE RESPONSE MONOTONIC? ANALYSIS OF INCOHERENT FEEDFORWARD MOTIFS *

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ABSTRACT

We study the monotonicity of the cumulative dose response (cDR) for a class of incoherent feedforward motif (IFFM) systems with linear intermediate dynamics and nonlinear output dynamics. While the instantaneous dose response (DR) may be nonmonotone with respect to the input, the cDR can still be monotone. To analyze this phenomenon, we derive an integral representation of the sensitivity of cDR with respect to the input and establish general sufficient conditions for both monotonicity and non-monotonicity. These results reduce the problem to verifying qualitative sign properties along system trajectories. We apply this framework to four canonical IFFM systems and obtain a complete characterization of their behavior. In particular, IFFM1 and IFFM3 exhibit monotone cDR despite potentially non-monotone DR, while IFFM2 is monotone already at the level of DR, which implies monotonicity of cDR. In contrast, IFFM4 violates these conditions, leading to a loss of monotonicity. Numerical simulations indicate that these properties persist beyond the structured initial conditions used in the analysis. Overall, our results provide a unified framework for understanding how network structure governs monotonicity in cumulative input–output responses.

Keywords First keyword · Second keyword · More

1 Introduction

Biological systems, from single cells to whole organisms, can adjust their responses to environmental signals. A key feature of this ability is maintaining internal stability under sustained inputs while still responding transiently to changes. This behavior, known as *adaptation* or *homeostasis*, arises from the interplay of signal transduction and gene regulation dynamics [1, 2].

A system exhibits *perfect adaptation* if its output returns to a steady state independently of the magnitude of a constant input. Although adaptation is an asymptotic property, many applications focus on finite-time behavior [3]. In particular, one studies how the output at a *fixed time* depends on the input. This functional dependency yields the notion of *dose response* (DR). The *cumulative dose response* (cDR), on the other hand, considers the total accumulated output over a time interval, which is often what experiments measure, the “area under the curve.” A key example of cDR arises in T-cell recognition, where immune cells must respond to antigens without overactivation [4], and the response is quantified by measuring accumulated cytokines over a finite period.

Incoherent feedforward motifs (IFFMs) define a class of network motifs known to generate adaptive responses. In these systems, the input affects the output both directly and indirectly through an intermediate regulatory pathway with an opposing effect [4], see Fig. 1a for an illustration. This antagonistic structure enables such motifs to exhibit adaptation and to detect changes in the input, see Fig. 1b and some studies in [5–7] using the simplest models.

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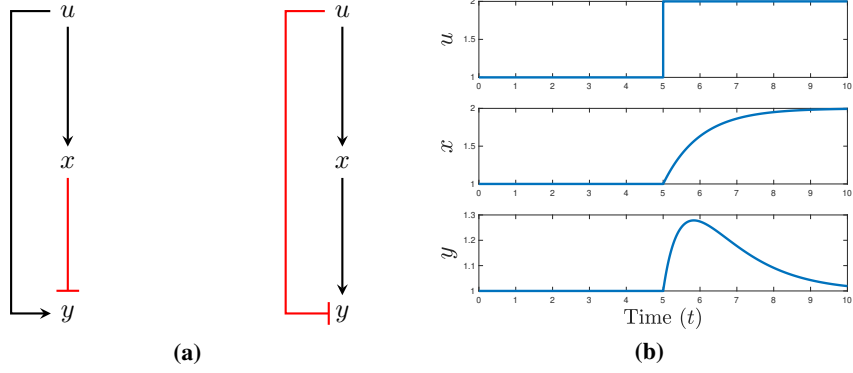


Figure 1: (a) IFFMs with inhibition via x (left) or directly from u (right). (b) An example: suppose that the external input u changes from a baseline value $u = 1$ to a new constant value $u = 2$ at time $5s$, and $x(0) = 1, y(0) = 1$ (steady state for $u = 1$ in system $\dot{x} = -x + u, \dot{y} = u/x - y$). This acts as a “change detector”: $\Delta u(t)$ triggers an activity burst, followed by a return to the adapted value, $y(t) \rightarrow y(0)$.

As explained in [4, 8], experimental work leads one to the following question: how do the DR and cDR, at a given finite time, depend on the magnitude of a constant stimulus input u ? Specifically, when is the cDR monotonic as a function of the input magnitude u ? Note that this is a *very different question from asking about monotonicity or not of the output as a function of time*. From now on, when we refer to monotonicity, we mean in this sense of dependence on input magnitude. An interesting phenomenon, discussed in the above papers, is that even if the DR happens to be nonmonotonic, the cDR could be monotonic, and this makes the analysis much harder. The study [8] established the monotonicity of cDR for two two-dimensional IFFMs, both having a linear first-order intermediate node dynamics. In this work, we investigate monotonicity properties of the cumulative dose response for a larger class of systems in which the intermediate node has n -dimensional linear dynamics, encompassing several canonical IFFM structures. Our main contributions are as follows:

- We formulate a general class of dynamical systems extending beyond first-order models, allowing for vector-valued intermediate states and nonlinear output dynamics.
- We derive an integral representation of the sensitivity of the cumulative dose response (cDR) with respect to the input. This leads to two general results: a sufficient condition for monotonicity and a complementary sufficient condition for non-monotonicity. Together, these results reduce the problem of determining monotonicity to verifying qualitative sign conditions along system trajectories.
- We apply this framework to four canonical IFFM systems and obtain a complete characterization of their monotonicity properties. In particular, we show that certain IFFM structures exhibit monotone cumulative responses despite nonmonotone dose responses, while others fail to satisfy the required conditions.
- We identify two novel mechanisms of IFFMs that lead to nonmonotonic response.

These results provide a more unified perspective on monotonicity of cumulative responses and clarify the role of network structure in shaping input–output behavior.

2 Theoretical Framework

In this paper, we let $\mathbb{R}_+^{n \times m} := \{A \in \mathbb{R}^{n \times m} \mid A > 0\}$ be the set of $n \times m$ matrices with positive entries. The sets $\mathbb{R}_+^n$ and $\mathbb{R}_+$ are defined similarly. The orders $>, <, \geq,$ and $\leq$ are interpreted elementwise when applied to vectors and matrices. The symbol $\mathbf{0}_p$ is a zero vector of dimension p .

2.1 Motivating model and assumptions

By definition, an incoherent feedforward motif (IFFM) is a system where the input u affects the output y both directly and indirectly through an intermediate state x . Here, x acts as a *delayed representation* of u , such that y receives *conflicting signals* (activation and repression). Before turning to more general dynamics to allow the modeling of possibly longer delays (or other filtering properties), we consider a simple model, in order to motivate the setting.

Table 1: Admissible scalar IFFM systems

No.	α_1	β_1	α_2	β_2	$\dot{y}$
1	1	-1	0	0	$\frac{x}{u} - y$
2	1	0	0	1	$x - uy$
3	-1	1	0	0	$\frac{u}{x} - y$
4	-1	0	0	-1	$\frac{1}{x} - \frac{1}{u}y$
5	0	1	1	0	$u - xy$
6	0	0	1	-1	$1 - \frac{x}{u}y$
7	0	-1	-1	0	$\frac{1}{u} - \frac{1}{x}y$
8	0	0	-1	1	$1 - \frac{u}{x}y$

We use a first-order stable linear dynamics $\dot{x} = -ax + bu$, $u \in \mathbb{R}_+$, as a model of a delay. For the y -dynamics receiving the conflicting signals, we make these modeling assumptions:

1. Both x and u act either as activators or inhibitors (but not both simultaneously), and no homodimers form².
2. The system is stable and exhibits perfect adaptation to constant inputs, meaning that the steady-state value of y is independent of x and u . Moreover, the response is not independent of u (finite relative degree).
3. Any Michaelis–Menten constants are sufficiently small and are approximated by zero. For instance, terms of the form $u/(K + x)$ are approximated by ux^{-1} (nonzero K will be allowed in our actual theorems).

Thus, we consider dynamics of the form:

$$\dot{y} = cx^{\alpha_1}u^{\beta_1} - dx^{\alpha_2}u^{\beta_2}y, \quad (1)$$

where the exponents $\alpha_i, \beta_i \in \{-1, 0, 1\}$. Here c, d are positive constants; for simplicity in this motivational discussion, we take $a = b = c = d = 1$. The above assumptions get reflected into the following structural constraints on the parameters:

1. Each of x and u appears exactly once: $|\alpha_1| + |\alpha_2| = 1$, and $|\beta_1| + |\beta_2| = 1$.
2. Equal degrees: $\alpha_1 + \beta_1 = \alpha_2 + \beta_2$.

The first property corresponds to the assumption that no variable can be both an activator and an inhibitor, and the second property guarantees perfect adaptation, since at steady state $y = u^{(\alpha_1 + \beta_1) - (\beta_1 + \beta_2)}$. The admissible exponent combinations are given in Table 1, in which the corresponding y equations are shown in the last column.

2.2 Incoherent feedforward motifs (IFFMs) dynamics

In general, motivated by the assumptions in the previous subsection, we consider systems of the form:

$$\dot{x}(t) = Ax(t) + bu, \quad \dot{y}(t) = F(x(t), y(t), u), \quad (2)$$

where $x(t) \in \mathbb{R}^n$ is the intermediate state, $y(t) \in \mathbb{R}$ is the output of the system, and the input $u \in \mathbb{R}_+$ is constant. We assume that the x -subsystem is stable positive³.

Assumption 1 *The x -subsystem in (2) is stable and positive, i.e., for all $x(0) \in \mathbb{R}_+^n$ and inputs $u(t) \in \mathbb{R}_+^n$, the corresponding solutions satisfy $x(t) \geq 0, \forall t \geq 0$. Equivalently, the matrix $A \in \mathbb{R}^{n \times n}$ is Hurwitz and Metzler, and $b \in \mathbb{R}_+^n$.*

The function $F : \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}_+ \rightarrow \mathbb{R}$ is of class C^1 . Its partial derivatives with respect to x and u have opposite signs, hence $\partial_u F(\cdot) \partial_x F(\cdot) < 0$, as in the examples in Table 1, representing the “incoherent” effects of x and u on the output. Moreover, motivated by the previous discussion of systems (1), in our main results we will restrict attention to systems

²There are no powers u^i or x^i with $i > 1$.

³The restriction to positive systems is natural in biology, where the states represent typically concentrations of molecules.

in which F has forms shown in the last column of Table 1, where instead of “ cx ” or “ dx ” we have a linear function $c^\top x$ or $d^\top x$ with a positive vector c or d (to preserve positivity of the linear system).

Unless otherwise stated, we consider initial conditions $x_0 \geq 0$ and $y_0 \in \mathbb{R}_+$. In particular, when convenient, we choose $y_0 = y_{ss}$, where y_{ss} denotes the steady state of the corresponding y -subsystem evaluated at the steady state $x_{ss}(u) := -A^{-1}bu \in \mathbb{R}_+^n$. For each fixed input $u \in \mathbb{R}_+$ and initial condition (x_0, y_0) , the system (2) admits a unique solution $(x_u(t), y_u(t))$ ⁴ defined for all $t \geq 0$. Moreover, the positive orthant is forward invariant, i.e., if $u \in \mathbb{R}_+$, $x_0 \geq 0$ and $y_0 \geq 0$, then $x_u(t) \geq 0$ and $y_u(t) \geq 0$ for all $t \geq 0$. The following remark is immediate.

Remark 1 Since A is Hurwitz, $x_u(t)$ converges exponentially to the equilibrium $x_{ss}(u) = -A^{-1}bu$. For each IFFM system, the y -subsystem preserves nonnegativity. In addition, since $x_u(t)$ is bounded, standard comparison arguments imply that $y_u(t)$ remains bounded for all $t \geq 0$.

2.3 Dose response and cumulative dose response

Recall $(x_u(t), y_u(t))$ as a unique solution of (2) for a fixed u . We define the dose response (DR) and cumulative dose response (cDR) at time $T > 0$ as

$$\text{DR}(u, T) := y_u(T), \quad \text{cDR}(u, T) := \int_0^T y_u(t) dt. \quad (3)$$

The dose response $\text{DR}(u, T)$ represents the instantaneous output at a fixed time T , while the cumulative dose response $\text{cDR}(u, T)$ captures the total accumulated effect over the time interval $[0, T]$.

It is important to note that monotonicity of $\text{DR}(u, T)$ is not guaranteed. In general, the map $u \mapsto \text{DR}(u, T)$ may be nonmonotone, because of the nonlinear dynamics of the y -subsystem. This nonmonotonicity reflects the fact that increasing the input can have competing effects over time, leading to peaks or nontrivial transient behavior.

However, we will show that nonmonotonicity of $\text{DR}(u, T)$ does not necessarily imply nonmonotonicity of $\text{cDR}(u, T)$: the cumulative response can remain monotone even when the instantaneous response is not. To study this problem, we analyze the sensitivity of $y_u(t)$ with respect to the input u and derive an integral representation of $\partial_u \text{cDR}(u, T)$. This representation will allow us to identify structural conditions under which monotonicity of cDR is guaranteed.

3 On the monotonicity of $\text{cDR}(u, T)$

In this part, we characterize the monotonicity of the map $u \mapsto \text{cDR}(u, T)$. First, recall the IFFM system in (2) and its solution $(x_u(t), y_u(t))$ defined on $[0, T]$, with initial condition (x_0, y_0) . Define the sensitivities:

$$p_u(t) := \partial_u x_u(t) \in \mathbb{R}^n \quad \text{and} \quad q_u(t) := \partial_u y_u(t) \in \mathbb{R}.$$

Then $q_u(t)$ satisfies

$$\dot{q}_u(t) + a_u(t)q_u(t) = g_u(t) \quad \text{and} \quad q_u(0) = 0, \quad (4)$$

where, evaluated along $(x_u(t), y_u(t), u)$,

$$a_u(t) := -\partial_y F(\cdot), \quad g_u(t) := \partial_x F(\cdot)p(t) + \partial_u F(\cdot).$$

Define further

$$G_u(t) := \int_0^t g_u(s) ds, \quad (5)$$

and

$$\lambda_u(t) := \int_t^T \exp\left(-\int_t^s a_u(\tau) d\tau\right) ds, \quad t \in [0, T]. \quad (6)$$

Notice that monotonicity of the map $u \mapsto \text{cDR}(u, T)$ is equivalent to requiring that $\partial_u \text{cDR}(u, T)$ has a fixed sign for all $T > 0$. Then, the following theorem provides sufficient conditions for this property in (2).

Theorem 1 Consider the system (2). Let $\text{cDR}(u, T)$, $g_u(t)$, $G_u(t)$, $\lambda_u(t)$ be defined as in (3)–(6). For $T > 0$, if either

- (i) each of $\lambda_u(t)$ and $g_u(t)$ has a fixed sign for $t \in [0, T]$, independent of u , or

⁴The dependence on the initial conditions will not be indicated explicitly; they will be fixed at values to be discussed.

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(ii) each of $\dot{\lambda}_u(t)$ and $G_u(t)$ has a fixed sign for $t \in [0, T]$, independent of u ,

then $\partial_u \text{cDR}(u, T)$ has a fixed sign. In particular, the map $u \mapsto \text{cDR}(u, T)$ is monotonic.

Proof 1 By the variation of parameters formula applied to (35),

$$q_u(t) = \int_0^t \exp\left(-\int_s^t a_u(\tau) d\tau\right) g_u(s) ds. \quad (7)$$

Since $q_u(t) = \partial_u y_u(t)$, integrating over $[0, T]$ and applying Fubini's theorem (changing the order of integration over $\{(s, t) : 0 \leq s \leq t \leq T\}$), we obtain

$$\begin{aligned} \partial_u \text{cDR}(u, T) &= \int_0^T \partial_u y_u(t) dt = \int_0^T q_u(t) dt \\ &= \int_0^T \int_s^T \exp\left(-\int_s^t a_u(\tau) d\tau\right) dt g_u(s) ds \\ &= \int_0^T \lambda_u(s) g_u(s) ds. \end{aligned} \quad (8)$$

We now consider the two cases using (8).

(i) If both $\lambda_u(t)$ and $g_u(t)$ have a fixed sign on $[0, T]$, then their product $\lambda_u(t)g_u(t)$ also has a fixed sign. Therefore, $\partial_u \text{cDR}(u, T)$ has a fixed sign.

(ii) Suppose that $\dot{\lambda}_u(t) \leq 0$ on $[0, T]$ and that $G_u(t)$ has a fixed sign on $[0, T]$. Since $g(t) = \dot{G}(t)$, integration by parts on (8) yields

$$\int_0^T \lambda_u(t) \dot{G}_u(t) dt = [\lambda_u(t) G_u(t)]_0^T - \int_0^T \dot{\lambda}_u(t) G_u(t) dt,$$

where the boundary term $[\lambda_u(t) G_u(t)]_0^T$ vanishes due to $G_u(0) = 0$ and $\lambda_u(T) = 0$. Hence, by assumption (ii), the integrand $-\dot{\lambda}_u(t) G_u(t)$ has a fixed sign on $[0, T]$. Thus, $\partial_u \text{cDR}(u, T)$ has a fixed sign.

In either case, $\partial_u \text{cDR}(u, T)$ has a fixed sign. Therefore, the map $u \mapsto \text{cDR}(u, T)$ is monotone.

Theorem 1 requires a sign structure that holds uniformly with respect to u in order to guarantee monotonicity of $u \mapsto \text{cDR}(u, T)$. Therefore, if this structure fails for some values of u , the map cannot be monotone. In particular, if $\partial_u \text{cDR}(u, T)$ takes opposite signs for two values of u , then monotonicity is violated. Therefore, we formalize this result in the following theorem.

Theorem 2 Consider the system (2). Let $\text{cDR}(u, T)$, $g_u(t)$, $G_u(t)$, $\lambda_u(t)$ be defined as in (3)–(6). Assume that there exist u_- , $u_+ > 0$ such that one of the following holds:

- (i) $\lambda_{u_-}(t)g_{u_-}(t) \geq 0$ and $\lambda_{u_-}(t)g_{u_-}(t) \not\equiv 0$, whereas $\lambda_{u_+}(t)g_{u_+}(t) \leq 0$ and $\lambda_{u_+}(t)g_{u_+}(t) \not\equiv 0$; or
- (ii) $-\dot{\lambda}_{u_-}(t)G_{u_-}(t) \geq 0$ and $-\dot{\lambda}_{u_-}(t)G_{u_-}(t) \not\equiv 0$, while $-\dot{\lambda}_{u_+}(t)G_{u_+}(t) \leq 0$ and $-\dot{\lambda}_{u_+}(t)G_{u_+}(t) \not\equiv 0$,

for all $t \in [0, T]$. Then

$$\partial_u \text{cDR}(u_-, T) > 0, \quad \partial_u \text{cDR}(u_+, T) < 0, \quad (9)$$

and the map $u \mapsto \text{cDR}(u, T)$ is not monotone on any interval containing both u_- and u_+ .

Proof 2 Recall (8). Thus, under assumption (i), the integrand has a fixed sign for $u = u_-$ and the opposite fixed sign for $u = u_+$, and is not identically zero in either case. Hence, proving (9). Alternatively, integrating by parts as in the proof of Theorem 1. Therefore, under assumption (ii), the integrand $-\dot{\lambda}_u G_u$ has one fixed sign for $u = u_-$ and the opposite fixed sign for $u = u_+$, and is not identically zero in either case. Hence again, proving (9). Thus $u \mapsto \text{cDR}(u, T)$ cannot be monotone.

Next, we leverage Theorems 1 and 2 to analyze the monotonicity of system (2), which generalizes the scalar IFFM systems listed in Table 1.

4 Applications to IFFM systems

Note that Theorem 1 reduces the monotonicity problem to the analysis of the sign of $g_u(t)$ and the associated kernel $\lambda_u(a(t))$. This reduction allows for a systematic analysis of the IFFM systems. In particular, for each system we proceed the algorithm/steps as follows⁵:

1. Compute $p_u = \partial_u x_u$ by solving $\dot{p}_u = Ap_u + b$.
2. Define $q_u = \partial_u y_u$ and differentiate to obtain (35) thereby identifying the functions a_u and g_u .
3. Compute λ_u and analyze its qualitative behavior, in particular its sign and monotonicity.
4. Analyze g_u and G_u , in order to determine its sign.

Applying these steps to the IFFM systems listed in Table 1, we obtain the summary in Table 2 (see Appendix B for the scalar case and Appendix C for the vector case). In the next result, we summarize the behavior of some systems.

Proposition 1 *Let $\dot{x} = -x + u$, $u \in \mathbb{R}_+$, and y -dynamics no. 1, 3, 6, and 8 listed in Table 1. By Theorem 1, those systems admit monotonic $\text{cDR}(u, T)$.*

Proof 3 *From Table 2, $a(t) \geq 0$ in those systems, therefore $\lambda_u(t) \geq 0, \forall t \in [0, T]$. Moreover, $g_u(t)$ has a fixed sign, hence $G_u(t)$. By Theorem 1 (i), those systems admit monotonic $\text{cDR}(u, T)$.*

We extend this analysis to the corresponding vector versions of cases 1, 3, 6, and 8 (see Appendix C). In the main body of the paper, we provide detailed analysis for cases 3, 2, 4, and 5. Case 7 is analogous to case 4 and is therefore treated in Appendix B and Appendix C. We remark that cases 3 and 5 were previously studied in [8], albeit only in the scalar setting. The proofs presented here are more streamlined and extend naturally to the vector case.

In the rest of this paper, we refer to systems 5, 3, 2, and 4 in their general vector form as IFFM1, IFFM2, IFFM3, and IFFM4, respectively.

Table 2: Summary of $a(t)$, $g(t)$, and $G(t)$ for scalar IFFM systems with $p(t) = 1 - e^{-t}$, $x(t) \geq 0$ and $y(t) \geq 0$ for all $t \geq 0$.

No.	System	$a(t)$	$g(t)$	$G(t)$
1	$\frac{x}{u} - y$	1	$-\frac{x_0 e^{-t}}{u^2} < 0$	$-\frac{x_0}{u^2}(1 - e^{-t}) < 0$
2	$x - uy$	u	$p - y$	$\int_0^t (p - y) ds$
3	$\frac{u}{x} - y$	1	$\frac{x_0 e^{-t}}{x^2} > 0$	$\int_0^t \frac{x_0 e^{-s}}{x(s)^2} ds > 0$
4	$\frac{1}{x} - \frac{1}{u}y$	$\frac{1}{u}$	$-\frac{p}{x^2} + \frac{y}{u^2}$	$\int_0^t \left(-\frac{p}{x^2} + \frac{y}{u^2}\right) ds$
5	$u - xy$	x	$1 - yp$	$t - \int_0^t yp ds$
6	$1 - \frac{x}{u}y$	$\frac{x}{u}$	$\frac{x_0 e^{-t}}{u^2}y > 0$	$\int_0^t \frac{x_0 e^{-s}}{u^2}y(s) ds$
7	$\frac{1}{u} - \frac{1}{x}y$	$\frac{1}{x}$	$\frac{yp}{x^2} - \frac{1}{u^2}$	$\int_0^t \left(\frac{yp}{x^2} - \frac{1}{u^2}\right) ds$
8	$1 - \frac{u}{x}y$	$\frac{u}{x}$	$-\frac{x_0 e^{-t}}{x^2}y < 0$	$-\int_0^t \frac{x_0 e^{-s}}{x(s)^2}y(s) ds$

4.1 IFFM1: Monotonicity of $\text{cDR}(u, T)$

Recall the IFFM1 system:

$$\dot{x}_u = Ax_u + bu, \quad \dot{y}_u = -c^\top x_u y_u + du, \quad (10)$$

⁵We drop the dependence on (t) when no confusion arises.

where $b, c \in \mathbb{R}_+^n$ and $d \in \mathbb{R}_+$. Notice that all IFFM systems share the same x -subsystem. Consequently, the sensitivity $p_u = \partial_u x_u$ is identical in all cases. Define $p_u := \partial_u x_u \in \mathbb{R}^n$. Differentiating with respect to u yields $\dot{p}_u = Ap_u + b$ with $p_u(0) = 0$. Hence, the solution is given by

$$p_u = \int_0^t e^{A(t-s)} b \, ds = A^{-1}(e^{At} - I)b. \quad (11)$$

Since A is Metzler, $e^{A(t-s)} \geq 0$ for all $t \geq s$, and since $b \in \mathbb{R}_+^n$, it follows that $p_u \geq 0$ for all $t \geq 0$. Next, define $q_u := \partial_u y_u \in \mathbb{R}$. Differentiating with respect to u yields

$$\dot{q}_u = -c^\top p_u y_u - c^\top x_u q_u + d, \quad q_u(0) = 0.$$

Thus, q_u satisfies (35) with $a_u = c^\top x_u$ and $g_u = d - c^\top p_u y_u$.

Regarding λ_u , since $c \in \mathbb{R}_+^n$ and $x_u \geq 0$ for all $t \geq 0$, we have $a_u = c^\top x_u \geq 0$. Therefore, the exponential factor in the definition of λ_u is strictly positive, and

$$\lambda_u = \int_t^T \exp\left(-\int_t^s c^\top x_u(\tau) \, d\tau\right) ds \geq 0, \quad \forall t \in [0, T].$$

Moreover, $\lambda(T) = 0$, and for every $t < T$ the interval $[t, T]$ has positive length, hence $\lambda_u > 0$ for all $t < T$. However, its detailed behavior depends on the trajectory x_u . In particular, the qualitative behavior of x_u depends on the initial condition x_0 relative to the steady state $x_{ss}(u)$. For this reason, at this stage we do not attempt to derive an explicit expression for λ_u or its derivative.

Furthermore, in the following Lemma⁶, we show that g_u and G_u remain nonnegative.

Lemma 1 *Consider system (10), where A is Metzler and Hurwitz, with $b, c \in \mathbb{R}_+^n$ and $d \in \mathbb{R}_+$. Fix $u \in \mathbb{R}_+$ and let (x_u, y_u) denote the corresponding solution. Assume that $x_0 = -A^{-1}bv$ for some $v \in [0, \infty)$ and $y_0 = y_{ss}$ with $y_{ss} = d/[c^\top(-A^{-1}b)]$. Define $g_u := d - c^\top p_u y_u$, where $p_u = \partial_u x_u$. Then the following hold for all $t \in [0, T]$:*

- if $v \geq u$, then $g_u \geq 0$;
- if $v < u$, then $G_u \geq 0$.

For the case $v \geq u$, we have already established monotonicity, and will revisit this case later. For the case $v < u$, we show below that λ_u is nonincreasing.

Proposition 2 *Assume $v < u$. Let λ_u be given by (6) with $a_u = c^\top x_u$. Then $\dot{\lambda}_u \leq 0$ for all $t \in [0, T]$.*

We now apply Theorem 1. If $v \geq u$, then by Lemma 1, $g_u \geq 0, \forall t \in [0, T]$. Since $\lambda_u \geq 0$, it follows immediately that $\partial_u \text{cDR}(u, T) \geq 0$. If $v < u$, By Lemma 1, $G_u \geq 0$ for all $t \in [0, T]$, and by Proposition 2, $\dot{\lambda}_u \leq 0$ on $[0, T]$. Using the integration by parts from the proof of Theorem 1,

$$\partial_u \text{cDR}(u, T) = -\int_0^T \dot{\lambda}_u G_u \, dt \geq 0,$$

in which the integrand $-\dot{\lambda}_u G_u$ has a fixed sign. Hence, $\partial_u \text{cDR}(u, T)$ has a fixed sign, and therefore the map $u \mapsto \text{cDR}(u, T)$ is monotone for system (10).

4.2 IFFM2: Monotonicity of $\text{cDR}(u, T)$

Recall the IFFM2 system:

$$\dot{x}_u = Ax_u + bu, \quad \dot{y}_u = \frac{\beta u}{K + c^\top x_u} - dy_u, \quad (12)$$

where $\beta, K \in \mathbb{R}_+$. We will show that for any fixed $T > 0$, the dose response $\text{DR}(u, T) = y_u(T)$ is monotone. This implies that the map $u \mapsto \text{cDR}(u, T)$ is also monotone. To this end, we show that $\forall t \geq 0$, the map $u \mapsto y_u$ is monotone.

⁶For simplicity and tractability of the analysis, we assume that the initial conditions satisfy $x_0 \in \{-A^{-1}bv \mid v \in [0, \infty)\}$, $y_0 = y_{ss}$. In simulation, we use general x_0 such that $x_0 \notin \text{span}\{-A^{-1}b\}$.

Monotonicity of cumulative dose response $\text{cDR}(u, T)$

Let $q_u = \partial_u y_u$. Differentiating with respect to u yields

$$\dot{q}_u = \frac{\beta}{K + c^\top x_u} - \frac{\beta u c^\top p_u}{(K + c^\top x_u)^2} - dq_u,$$

with $q_u(0) = 0$. Using $c^\top x_u = c^\top e^{At} x_0 + u c^\top p_u$, we obtain the expression of $c^\top p_u$. Substituting into $\dot{q}_u$ yields

$$\begin{aligned} \dot{q}_u &= \frac{\beta}{K + c^\top x_u} - \frac{\beta(c^\top x_u - c^\top e^{At} x_0)}{(K + c^\top x_u)^2} - dq_u \\ &= \frac{\beta(K + c^\top e^{At} x_0)}{(K + c^\top x_u)^2} - dq_u. \end{aligned} \quad (13)$$

Since $K \in \mathbb{R}_+$, $e^{At} \geq 0$, $c \in \mathbb{R}_+^n$, and $x_0 \geq 0$, we have $\dot{q}_u \geq -dq_u$. With $q_u(0) = 0$, the comparison principle gives $q_u \geq 0, \forall t \geq 0$. Therefore q_u is nondecreasing and hence the map $u \mapsto y_u$ is monotone. This also concludes that the map $u \mapsto \text{cDR}(u, T)$ is also monotone for system (12).

4.3 IFFM3: Monotonicity of $\text{cDR}(u, T)$

Recall the IFFM3 system:

$$\dot{x}_u = Ax_u + bu, \quad \dot{y}_u = c^\top x_u - d y_u. \quad (14)$$

Let $q_u = \partial_u y_u$. Differentiating with respect to u yields

$$\dot{q}_u = c^\top p_u - d y_u - d u q_u, \quad q_u(0) = 0.$$

Thus, q_u satisfies (35) with $a_u = du$ and $\tilde{g}_u = c^\top p_u - d y_u$. In this case, λ_u admits an explicit expression

$$\lambda_u = \int_t^T e^{-du(\tau-t)} d\tau = \frac{1 - e^{-du(T-t)}}{du}. \quad (15)$$

Hence $\lambda_u \geq 0$ on $[0, T]$, $\lambda(T) = 0$. Since the derivative of (19) is $\dot{\lambda}_u = -e^{-du(T-t)}$ and $e^{-du(T-t)} > 0$ for all $t \in [0, T]$, it follows that $\dot{\lambda}_u < 0$ for all $t \in [0, T]$. Thus, λ_u is strictly decreasing on $[0, T]$.

For convenient, let $g_u = -\tilde{g}_u = d y_u - c^\top p_u$. Using $p_u(0) = 0$ and $y_0 = y_{ss}$, then $g_u(0) = d y_{ss} = c^\top (-A^{-1}b) \geq 0$. Differentiate g_u and use $x_u = e^{At} x_0 + u p_u$ yield

$$\begin{aligned} \dot{g}_u &= d(c^\top x_u - d y_u) - c^\top \dot{p}_u \\ \dot{g}_u + d g_u &= d c^\top x_u - c^\top \dot{p}_u - d u c^\top p_u \\ &= d c^\top e^{At} x_0 - c^\top \dot{p}_u. \end{aligned} \quad (16)$$

The next lemma shows that the function g_u by solving (16) has nonnegative accumulated mass over every time horizon.

Lemma 2 Consider system (14), where A is Metzler and Hurwitz, with $b, c \in \mathbb{R}_+^n$ and $d \in \mathbb{R}_+$. Fix $u \in \mathbb{R}_+$ and let (x_u, y_u) denote the corresponding solution with $x_0 \geq 0$ and $y_0 = y_{ss}$ with $y_{ss} = c^\top (-A^{-1}b)/d$. Define $g_u := d y_u - c^\top p_u$, where $p_u = \partial_u x_u$. Then, for every $t \in [0, T]$, $G_u \geq 0$.

Finally, recall that $\tilde{g}_u = -g_u = c^\top p_u - d y_u$, such that $G_u \leq 0$ on $[0, T]$. Since $\dot{\lambda}_u = -e^{-du(T-t)} < 0$, we have $-\dot{\lambda}_u > 0$. By Theorem 1,

$$\partial_u \text{cDR}(u, T) = - \int_0^T \dot{\lambda}_u G_u dt \leq 0.$$

the map $u \mapsto \text{cDR}(u, T)$ is also monotone for system (14).

4.4 IFFM4: Nonmonotonicity of $\text{cDR}(u, T)$

Recall the IFFM4 system:

$$\dot{x}_u = Ax_u + bu, \quad \dot{y}_u = \frac{1}{K + c^\top x_u} - \frac{d y_u}{\beta u}, \quad (17)$$

Let $q_u = \partial_u y_u$. Differentiating with respect to u yields

$$\dot{q}_u = -\frac{c^\top p_u}{(K + c^\top x_u)^2} + \frac{d y_u}{\beta u^2} - \frac{d}{\beta u} q_u, \quad q_u(0) = 0.$$

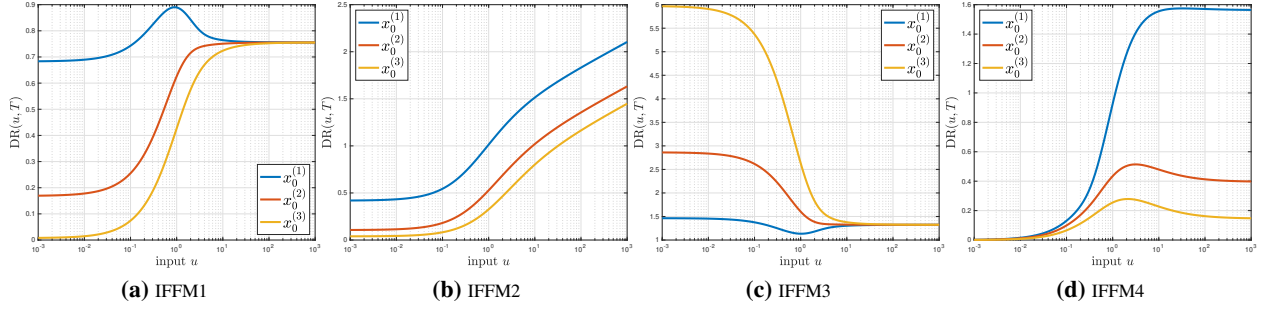


Figure 2: Dose response $DR(u, T)$ for the four IFFM systems under three different initial conditions x_0, y_0 . The input u is varied over $u \in [10^{-3}, 10^3]$. IFFM1 and IFFM3 exhibit nonmonotone DR, while IFFM2 shows monotone behavior. IFFM4 displays nonmonotonicity.

Thus, q_u satisfies (35) with $a_u = d/(\beta u)$ and

$$g_u = -\frac{c^\top p_u}{(K + c^\top x_u)^2} + \frac{dy_u}{\beta u^2}. \quad (18)$$

In this case, λ_u admits an explicit expression

$$\lambda_u = \int_t^T e^{-d(\tau-t)/\beta u} d\tau = \frac{\beta u}{d} (1 - e^{-d(T-t)/\beta u}). \quad (19)$$

Hence $\lambda_u \geq 0$ on $[0, T]$, $\lambda(T) = 0$. Since the derivative of (19) is $\dot{\lambda}_u = -e^{-d(T-t)/\beta u} < 0$ for all $t \in [0, T]$. Thus, λ_u is strictly decreasing on $[0, T]$.

The following lemma, we will show that the map $u \mapsto cDR(u, T)$ is not monotone using Theorem 2.

Lemma 3 Consider system (17), where A is Metzler and Hurwitz, with $b, c \in \mathbb{R}_+^n$ and $d, \beta, K \in \mathbb{R}_+$. Fix $u \in \mathbb{R}_+$ and let (x_u, y_u) denote the corresponding solution. Assume that $x_0 = 0$ and $y_0 = y_{ss} = \beta/[d(K + c^\top x_0)]$. Define g_u, G_u and $cDR(u, T)$ as in (18), (5), and (3). Then there exist $u_-, u_+ > 0$ such that

- $\partial_u cDR(u, T) > 0$ for all $0 < u < u_-$, and
- $\partial_u cDR(u, T) < 0$ for all $u > u_+$.

In particular, the map $u \mapsto cDR(u, T)$ is not monotone.

Having analyzed the monotonicity of $cDR(u, T)$ for the four generalized IFFM systems, we conclude that, except for systems 4 and 7 in Table 2, the behavior of $cDR(u, T)$ can be fully characterized using Theorem 1. In the next section, we complement this analysis with numerical simulations for the four generalized IFFM systems, which further illustrate and validate the theoretical results beyond initial condition assumptions in some of the proofs.

5 Numerical Results

In this section, we illustrate the theoretical results on monotonicity of the cumulative dose response (cDR) using four IFFM systems initialized in three different initial conditions (x_0, y_0) . The dose response (DR) and cumulative dose response (cDR) are defined as (3). For each system, we fix the time horizon $T = 1.5$, the initial conditions, and sweep the input over $u \in [10^{-3}, 10^3]$, sampled logarithmically. For the four variants, denoted IFFM1–IFFM4, all sharing the same x -subsystem $\dot{x}_u = Ax_u + bu$, where

$$A = \begin{bmatrix} -3.0 & 0.8 & 0.0 & 0.0 & 0.0 \\ 0.4 & -2.6 & 0.7 & 0.0 & 0.0 \\ 0.0 & 0.5 & -2.8 & 0.6 & 0.0 \\ 0.0 & 0.0 & 0.4 & -2.3 & 0.7 \\ 0.0 & 0.0 & 0.0 & 0.3 & -1.7 \end{bmatrix}, \quad b = \begin{bmatrix} 1.0 \\ 0.8 \\ 0.9 \\ 0.7 \\ 0.6 \end{bmatrix}, \quad c = \begin{bmatrix} 0.9 \\ 0.7 \\ 0.8 \\ 0.6 \\ 0.5 \end{bmatrix},$$

with A Metzler and Hurwitz, $b, c \geq 0$, and $d = 1.2$. For IFFM3 and IFFM4, we also use $\beta = 1.5$, $\gamma = 0.8$, and $K = 0.8$. Here, we consider more general initial conditions (x_0, y_0) , which allows us to test the robustness of the monotonicity

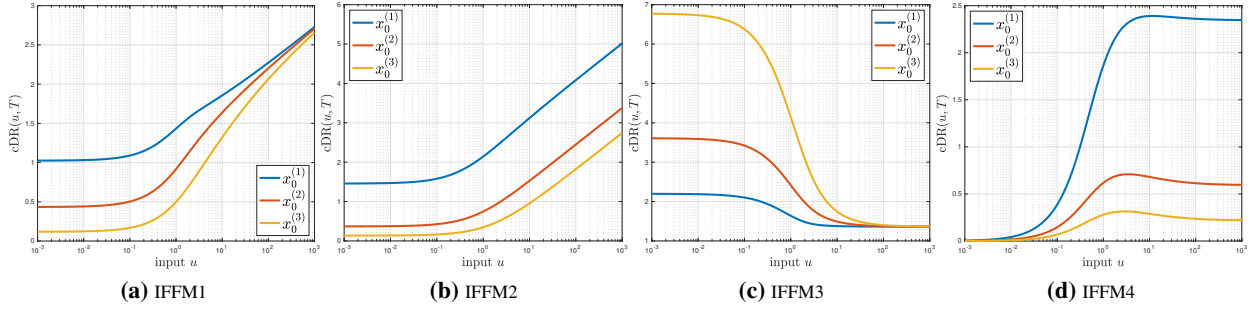


Figure 3: Cumulative dose response $cDR(u, T)$ for the four IFFM systems under the same conditions as Figure 2. Despite nonmonotone DR, IFFM1 and IFFM3 exhibit monotone cDR, in agreement with the theoretical results. IFFM2 remains monotone, while IFFM4 loses monotonicity due to the sign-indefiniteness of the sensitivity function G_u .

properties beyond the structured class used in the proofs, i.e. $x_0 \in \{-A^{-1}bv, v > 0\}$. For the three x_0 , we design $x_0^{(1)} = \mathbf{0}_5$, $x_0^{(2)} = [0.5, 0.6, 0.7, 0.8, 0.9]^\top$, and also $x_0^{(3)} = [2.0, 2.1, 2.3, 2.4, 2.5]^\top$ while for the three y_0 , we have the following. For IFFM1 and IFFM3, we set $y_0 = y_{ss}$ while for IFFM2 and IFFM4 we use $y_0 = \beta/[d(K + c^\top x_0)]$. Finally, the DR and cDR curves are shown in Figures 2 and 3, respectively.

As for IFFM1 and IFFM3, Figures 2a and 2c show that the dose response $DR(u, T)$ is *nonmonotone* with respect to the input u for some initial conditions. However, the corresponding cumulative responses $cDR(u, T)$ in Figures 3a and 3c are *monotone*. This is fully consistent with the theoretical results: although the instantaneous sensitivity g_u may change sign, the weighted integral in the expression of $\partial_u cDR$ remains nonnegative due to the positivity and monotonicity properties established in Lemma 1–2 and its proposition.

For IFFM2, in Figures 2b and 3b, both DR and cDR are monotone increasing. This reflects the stronger structural property of this configuration, where the sensitivity term $\partial_u y_u$ preserves a fixed sign. In particular, the corresponding g_u remains nonnegative, in agreement with the analytical conditions ensuring monotonicity at both instantaneous and cumulative levels.

Regarding IFFM4, in contrast, Figures 2d and 3d show that both DR and cDR are *nonmonotone*. This confirms the analysis of Theorem 2 and Lemma 3: the function g_u does not have a fixed sign, and its integral also changes sign, even with $T = 1.5$. Consequently, the monotonicity of cDR is lost. This provides a clear counterexample demonstrating that the structural conditions identified for IFFM1–IFFM3 are not satisfied in this case. For IFFM2 and IFFM4, the simulation for both $DR(u, T)$ and $cDR(u, T)$ do not go to steady state y_{ss} because we set $K \neq 0$.

Table 3: Monotonicity properties of DR and cDR

System	DR	cDR	a_u	G_u
IFFM1	Nonmonotone	Monotone	≥ 0	≥ 0
IFFM2	Monotone	Monotone	≥ 0	≥ 0
IFFM3	Nonmonotone	Monotone	≥ 0	≤ 0
IFFM4	Nonmonotone	Nonmonotone	≥ 0	No fixed sign

The behaviors observed in the simulations are summarized in Table 3. These numerical experiments corroborate the theoretical results and highlight the sharpness of the conditions derived in the previous sections. In particular, we show that monotonicity of cDR is robust under general initial conditions for IFFM1–IFFM3, but can fail when the structural conditions on the y -dynamics are violated, as in IFFM4.

6 Conclusion

We studied monotonicity properties of the cumulative dose response (cDR) for a class of incoherent feedforward motif (IFFM) systems. We first derived an integral representation of $\partial_u cDR(u, T)$ and showed that the monotonicity problem can be reduced to verifying sign conditions on a sensitivity term and its associated kernel along system trajectories. This led to a general sufficient condition for monotonicity (Theorem 1) and a complementary sufficient condition for non-monotonicity (Theorem 2), providing a unified framework to characterize the input–output behavior of these systems. We then applied this framework to four canonical IFFM systems. Although the dose response (DR) may

be non-monotone, we showed that for IFFM1 and IFFM3 the cDR is monotone, as the required sign conditions are satisfied. For IFFM2, monotonicity holds already at the level of DR and therefore also for cDR . In contrast, IFFM4 violates these conditions and, by Theorem 2, necessarily exhibits non-monotonic behavior for cDR . These results demonstrate that the proposed conditions are not only sufficient but, within this class of systems, essentially sharp in distinguishing monotone from non-monotone cumulative responses.

Numerical simulations indicate that these properties persist beyond the restricted class of initial conditions used in the analysis. Future work includes extending the framework to broader classes of nonlinear systems, relaxing positivity assumptions, and identifying minimal structural conditions under which monotonicity of cDR can be guaranteed.

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A Proofs

Proof of Lemma 1

First, we analyze the relation of x_u and y_u with respect to $x_{ss}(u)$ and y_{ss} respectively under two conditions: $v \geq u$ and $v < u$.

We start with x_u . Since $x_0 = -A^{-1}bv = x_{ss}(v)$, the solution of the x -subsystem can be written as

$$x_u = x_0 + p_u(u - v).$$

From (33), we have $\dot{p}_u = Ap_u + b = e^{At}b \geq 0, \forall t \geq 0$. Therefore, $\dot{x}_u = \dot{p}_u(u - v) = e^{At}b(u - v)$. It follows that x_u is nonincreasing when $v \geq u$, and nondecreasing when $v < u$. In particular,

$$x_u \in \begin{cases} [x_{ss}(u), x_0], & \text{if } v \geq u, \\ [x_0, x_{ss}(u)], & \text{if } v < u, \end{cases} \quad \forall t \geq 0.$$

Next, consider the y -subsystem in (10). Then, evaluating at $y_u = y_{ss}$ and using $du = y_{ss} c^\top x_{ss}(u)$, we obtain

$$\dot{y}_u|_{y_u=y_{ss}} = y_{ss} c^\top (x_{ss}(u) - x_u).$$

If $v \geq u$, then $x_u \geq x_{ss}(u)$, yielding $\dot{y}_u|_{y_u=y_{ss}} \leq 0$. Since $y_0 = y_{ss}$, the comparison principle yields $y_u \leq y_{ss}, \forall t \geq 0$. Similarly, if $v < u$, then $x_u \leq x_{ss}(u)$, so $\dot{y}_u|_{y_u=y_{ss}} \geq 0$, and hence $y_u \geq y_{ss}, \forall t \geq 0$.

Now, using the behaviors of x_u and y_u with respect to $x_{ss}(u)$ and y_{ss} respectively, we will prove the claims.

- **Case $v \geq u$.** Since $p_u \leq -A^{-1}b$ and $c \in \mathbb{R}_+^n$, we have $c^\top p_u \leq c^\top (-A^{-1}b)$. Together with $y_u \leq y_{ss}$, this yields $c^\top p_u y_u \leq c^\top (-A^{-1}b) y_{ss} = d$. Hence,

$$g_u = d - c^\top p_u y_u \geq 0, \quad \forall t \geq 0.$$

- **Case $v < u$.** Using $x_u = e^{As}x_0 + up_u$, we obtain $c^\top x_u = c^\top e^{As}x_0 + u c^\top p_u$. Substituting this into the y -subsystem and rearranging gives

$$g_u = d - c^\top p_u y_u = \frac{1}{u} \dot{y}_u + \frac{1}{u} (c^\top e^{As}x_0) y_u.$$

Integrating over $[0, t]$ and using $y_0 = y_{ss}$, we obtain

$$G_u = \frac{1}{u} (y_u - y_{ss}) + \frac{1}{u} \int_0^t (c^\top e^{As}x_0) y_u ds.$$

Since $x_0 \geq 0$, $e^{As} \geq 0$, $c \in \mathbb{R}_+^n$, and $y_u \geq 0$, the integral term is nonnegative. Moreover, $y_u \geq y_{ss}$, hence $G_u \geq 0$ for all $t \in [0, T]$.

This completes the proof.

Proof of Proposition 2

Differentiating λ_u gives $\dot{\lambda}_u = -1 + c^\top x_u \lambda_u$. By the proof of Lemma 1 for $v < u$, x_u is nondecreasing. Hence for every $s \in [t, T]$, $c^\top x_u(\tau) \geq c^\top x_u$, $\forall \tau \in [t, s]$, and therefore

$$\exp\left(-\int_t^s c^\top x_u(\tau) d\tau\right) \leq \exp(-(s-t)c^\top x_u).$$

Integrating from t to T and multiplying by $c^\top x_u \geq 0$, it follows that

$$\begin{aligned} c^\top x_u \lambda_u &\leq c^\top x_u \int_t^T \exp(-(s-t)c^\top x_u) ds \\ &= 1 - e^{-(T-t)c^\top x_u} \leq 1. \end{aligned}$$

Hence, $\dot{\lambda}_u = -1 + c^\top x_u \lambda_u \leq 0$, as claimed.

Proof of Lemma 2

Recall $g_u = dy_u - c^\top p_u$ and its derivative in (16) and let $f_u := dc^\top e^{At}x_0 - c^\top \dot{p}_u$. From (33), it follows that $\dot{p}_u = e^{At}b$, yielding $f_u = c^\top e^{At}(dx_0 - b)$, for all $t \geq 0$. The solution of (16) is

$$g_u = e^{-dut} g_u(0) + \int_0^t e^{-du(t-s)} f_u ds.$$

Integrating over $t \in [0, T]$ and exchanging the order of integration on $\{(s, t) : 0 \leq s \leq t \leq T\}$ yields

$$\int_0^T g_u dt = \lambda_u(0) g_u(0) + \int_0^T \lambda_u f_u ds. \quad (20)$$

Define

$$H(T) := \int_0^T \lambda_u e^{As} ds.$$

Since $e^{As} \geq 0$ for all $s \geq 0$, and $\lambda_u \geq 0$, it follows that $H \geq 0$. Substituting $f_u = c^\top e^{As}(dx_0 - b)$ into (20) gives

$$\begin{aligned} \int_0^T g_u dt &= \lambda_u(0) g_u(0) + c^\top H(dx_0 - b) \\ &= c^\top H(dx_0) + \lambda_u(0) g_u(0) - c^\top Hb. \end{aligned} \quad (21)$$

The first term $c^\top H(dx_0)$ is nonnegative because $c, x_0 \geq 0$ and $d > 0$. It remains to show $\lambda_u(0) g_u(0) - c^\top Hb \geq 0$. Since $\lambda_u \geq 0$ and λ_u is strictly decreasing on $[0, T]$, then $\lambda_u \leq \lambda_u(0)$ for all $s \in [0, T]$. Thus, with $b \in \mathbb{R}_+^n$ we get

$$\begin{aligned} Hb &= \int_0^T \lambda_u e^{As} b ds \leq \lambda_u(0) \int_0^T e^{As} b ds \\ &\leq \lambda_u(0) \int_0^\infty e^{As} b ds = -\lambda_u(0) A^{-1} b. \end{aligned}$$

Multiplying by $c^\top$ yields $c^\top Hb \leq \lambda_u(0) c^\top (-A^{-1}b)$. Since $g_u(0) = dy_{ss} = c^\top (-A^{-1}b)$, then $c^\top Hb \leq \lambda_u(0) g_u(0)$, i.e. $\lambda_u(0) g_u(0) - c^\top Hb \geq 0$. Substituting this into (21) results in $G_u \geq 0$ for $t \in [0, T]$, completing the proof.

Proof of Lemma 3

Recall $\dot{p}_u = Ap_u + b, p_u(0) = 0$. Let $r_u := c^\top p_u \geq 0$ with $r_u(0) = 0$ so that $\dot{r}_u(0) = c^\top b > 0$. Since $x_0 = 0$, we also have $c^\top x_u = ur_u$.

Recall $-\dot{\lambda}_u = e^{-d(T-t)/\beta u} > 0$ on $[0, T]$. Using $\dot{y}_u$ in (17), we then rewrite (18) as $g_u = -\dot{y}_u/u + K/[u(K + ur)^2]$. Integrating from 0 to t gives

$$G_u = \frac{y_0 - y_u}{u} + \frac{K}{u} \int_0^t \frac{ds}{(K + ur(s))^2}. \quad (22)$$

We now analyze the sign of $\partial_u \text{cDR}(u, T)$ through

$$\partial_u \text{cDR}(u, T) = \int_0^T e^{-d(T-t)/\beta u} G_u(t) dt.$$

Case: small u . If $u < dKy_0/\beta$, then for every $t \in [0, T]$,

$$\dot{y}_u|_{y_u=y_0} = \frac{1}{K + ur(t)} - \frac{dy_0}{\beta u} \leq \frac{1}{K} - \frac{dy_0}{\beta u} < 0.$$

Since $y_u(0) = y_0$, the comparison principle yields $y_u \leq y_0, t \in [0, T]$. Hence, by (22), $G_u \geq 0, t \in [0, T]$. Moreover, $G_u \not\equiv 0$ because the second term in (22) is strictly positive. Since $-\dot{\lambda}_u > 0$, Theorem 1 implies $\partial_u \text{cDR}(u, T) > 0$ for all sufficiently small $u > 0$.

Case: large u . Using variation of parameters formula on $\dot{y}_u$ and subtracting from y_0 , we can rewrite G_u in (22) as

$$\begin{aligned} G_u &= \frac{y_0(1 - e^{-dt/\beta u})}{u} - \frac{1}{u} \int_0^t \frac{e^{-d(T-t)/\beta u}}{K + ur(s)} ds \\ &\quad + \frac{K}{u} \int_0^t \frac{ds}{(K + ur(s))^2}. \end{aligned}$$

Since $\dot{r}_u(0) = c^\top b > 0$, then $r(s) \sim (c^\top b)s$ as $s \rightarrow 0$. It follows that, for each fixed $t > 0$,

$$\int_0^t \frac{e^{-d(T-t)/\beta u}}{K + ur(s)} ds \sim \frac{\log u}{u} \cdot \frac{1}{c^\top b}, \quad u \rightarrow \infty,$$

whereas $y_0(1 - e^{-dt/\beta u})/u = O(u^{-2})$ and similarly

$$\frac{K}{u} \int_0^t \frac{ds}{(K + ur(s))^2} = O(u^{-2}).$$

Therefore

$$G_u(t) = -\frac{\log u}{u^2} \cdot \frac{1}{c^\top b} + O(u^{-2}), \quad u \rightarrow \infty.$$

In particular, for every fixed $t > 0$, one has $G_u(t) < 0$ for all sufficiently large u . Thus, $\partial_u \text{cDR}(u, T) > 0$ for small u and $\partial_u \text{cDR}(u, T) < 0$ for large u , and therefore $u \mapsto \text{cDR}(u, T)$ is not monotone.

B Monotonicity of $\text{cDR}(u, T)$ for the scalar IFFM systems

In this subsection, we compute the quantities $a_u(t)$, $g_u(t)$, and

$$G_u(t) := \int_0^t g_u(s) ds$$

for the scalar IFFM systems.

For all systems, we consider $\dot{x}_u = -x_u + u$, with initial condition $x_u(0) = x_0$, independent of u . Differentiating it with respect to u gives $\dot{p}_u = -p_u + 1$ with $p_u(0) = 0$ where $p_u := \partial_u x_u$. Hence

$$p_u = 1 - e^{-t}, \quad \text{and} \quad x_u = u + (x_0 - u)e^{-t} \quad (23)$$

Let $q_u := \partial_u y_u$. If $\dot{y}_u = F(x_u, y_u, u)$, then differentiating with respect to u yields

$$\dot{q}_u = \partial_x F(x_u, y_u, u) p_u + \partial_y F(x_u, y_u, u) q_u + \partial_u F(x_u, y_u, u). \quad (24)$$

Rewriting (24) in the form $\dot{q}_u + a_u(t)q_u = g_u(t)$, we identify

$$a_u(t) = -\partial_y F(x_u, y_u, u), \quad g_u(t) = \partial_x F(x_u, y_u, u) p_u + \partial_u F(x_u, y_u, u).$$

System 1: $\dot{y}_u = \frac{x_u}{u} - y_u$. Here $F(x_u, y_u, u) = (x_u/u) - y_u$. Thus $\partial_x F = 1/u$, $\partial_y F = -1$, and $\partial_u F = -x_u/u^2$.

Therefore, $a_u = 1$ and

$$g_u = \frac{up_u - x_u}{u^2} = \frac{u(1 - e^{-t}) - u - (x_0 - u)e^{-t}}{u^2} = \frac{-x_0 e^{-t}}{u^2} \leq 0. \quad (25a)$$

Hence

$$G_u = -\frac{x_0}{u^2} \int_0^t e^{-s} ds = -\frac{x_0}{u^2} (1 - e^{-t}) \leq 0. \quad (25b)$$

Since $a_u = 1$ is constant, the kernel λ_u defined in Theorem 1 is

$$\lambda_u = \int_t^T e^{-(s-t)} ds = 1 - e^{-(T-t)}. \quad (25c)$$

In particular, $\lambda_u \geq 0$ for all $t \in [0, T]$, and $\dot{\lambda}_u = -e^{-(T-t)} < 0, \forall t \in [0, T]$.

Proposition 3 Consider the scalar System 1: $\dot{x}_u(t) = -x_u(t) + u$ and $\dot{y}_u(t) = (x_u(t)/u) - y_u(t)$ with arbitrary $x_0 \geq 0$ and $y_0 = y_{ss} = 1$. Then, for every $T > 0$, the $\text{cDR}(u, T)$ is monotone nonincreasing with respect to u .

Proof 4 The sensitivity $q_u(t) = \partial_u y_u(t)$ satisfies $\dot{q}_u(t) + q_u(t) = g_u(t)$ where $g_u(t)$ is given in (25a). From (25c), $\lambda_u(t) \geq 0$ for all $t \in [0, T]$, and $g_u(t) \leq 0$ for all $t \geq 0$. Thus $\lambda_u(t)g_u(t) \leq 0, \forall t \in [0, T]$. By Theorem 1 (i), $\partial_u \text{cDR}(u, T) \leq 0$. Hence $u \mapsto \text{cDR}(u, T)$ is monotone nonincreasing.

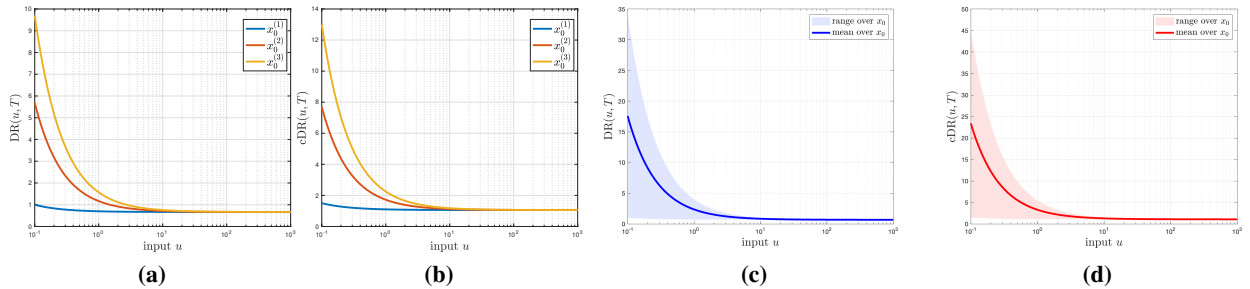


Figure 4: System 1: Dose response $\text{DR}(u, T)$ and cumulative dose response $\text{cDR}(u, T)$ for $u \in [10^{-1}, 10^3]$. (a)–(b) Responses for three fixed initial conditions x_0 . (c)–(d) Envelope (min–max band) and mean over $x_0 \in [0.1, 10]$. The results illustrate that $\text{cDR}(u, T)$ is monotone nonincreasing with respect to u , consistently with the theoretical analysis.

System 2: $\dot{y}_u = x_u - uy_u$. Here $F(x_u, y_u, u) = x_u - uy_u$. Thus $\partial_x F = 1$, $\partial_y F = -u$, and $\partial_u F = -y_u$.

Therefore, $a_u = u$ and $g_u = p_u - y_u$. Since $a_u = u$ is constant, the kernel λ_u defined in Theorem 1 is

$$\lambda_u = \int_t^T e^{-u(s-t)} ds = \frac{1 - e^{-u(T-t)}}{u}. \quad (26a)$$

In particular, $\lambda_u \geq 0$ for all $t \in [0, T]$, and $\dot{\lambda}_u = -e^{-u(T-t)} < 0, \forall t \in [0, T]$.

We now show that $G_u \leq 0$ for all $t \in [0, T]$, assuming $y_0 = y_{ss} = 1$ and arbitrary $x_0 \geq 0$. Using (23) and $\dot{y}_u = x_u - uy_u$, we rewrite g_u as

$$g_u = p_u - y_u = \frac{1}{u} \dot{y}_u - \frac{x_0 e^{-t}}{u}. \quad (26b)$$

Integrating (26b) over $[0, t]$, we obtain

$$G_u = \frac{1}{u} (y_u - y_0) - \frac{x_0}{u} (1 - e^{-t}). \quad (26c)$$

Next, using x_u in (23) and $y_0 = 1$, the y -subsystem may be written as $\dot{y}_u + u(y_u - 1) = (x_0 - u)e^{-t}$ with $y_u(0) - 1 = 0$. By variation of parameters with integrating factor e^{ut} ,

$$y_u - 1 = (x_0 - u) \int_0^t e^{-u(t-s)} e^{-s} ds. \quad (26d)$$

Substituting (26d) into (26c) gives

$$\begin{aligned} G_u &= \frac{x_0 - u}{u} \int_0^t e^{-u(t-s)} e^{-s} ds - \frac{x_0}{u} \int_0^t e^{-s} ds \\ &= - \int_0^t e^{-u(t-s)} e^{-s} ds + \frac{x_0}{u} \int_0^t (e^{-u(t-s)} - 1) e^{-s} ds. \end{aligned} \quad (26e)$$

Since $e^{-u(t-s)} \leq 1$ for all $0 \leq s \leq t$, both terms in (26e) are nonpositive. Hence $G_u \leq 0, \forall t \in [0, T]$.

Proposition 4 Consider the System 2 $\dot{x}_u(t) = -x_u(t) + u$ and $\dot{y}_u(t) = x_u(t) - uy_u(t)$ with arbitrary $x_0 \geq 0$ and $y_0 = y_{ss} = 1$. Then, for every $T > 0$, the $\text{cDR}(u, T)$ is monotone nonincreasing with respect to u .

Proof 5 The sensitivity $q_u(t) = \partial_u y_u(t)$ satisfies $\dot{q}_u(t) + uq_u(t) = g_u(t)$ where $g_u(t)$ is given by (26b). From (26a) we have $-\dot{\lambda}_u(t) > 0$ while by (26e), $G_u(t) \leq 0$ for all $t \in [0, T]$ and all $x_0 \geq 0$. Thus $-\dot{\lambda}_u(t) G_u(t) \leq 0, \forall t \in [0, T]$. By Theorem 1 (ii), $\partial_u \text{cDR}(u, T) \leq 0$. Hence $u \mapsto \text{cDR}(u, T)$ is monotone nonincreasing.

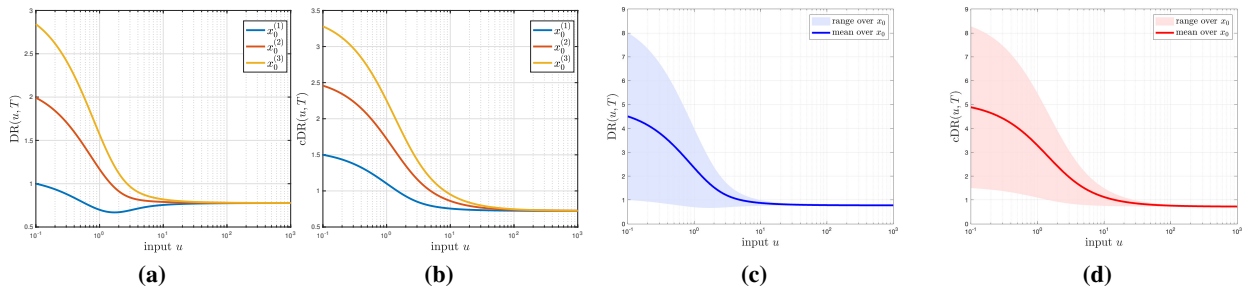


Figure 5: System 2: Dose response $\text{DR}(u, T)$ and cumulative dose response $\text{cDR}(u, T)$ for $u \in [10^{-1}, 10^3]$. (a)–(b) Responses for three fixed initial conditions x_0 . (c)–(d) Envelope (min–max band) and mean over $x_0 \in [0.1, 10]$. The results show that $\text{cDR}(u, T)$ is monotone nonincreasing with respect to u .

System 3: $\dot{y}_u = \frac{u}{x_u} - y_u$. Here $F(x_u, y_u, u) = (u/x_u) - y_u$. Thus $\partial_x F = -u/x_u^2$, $\partial_y F = -1$, and $\partial_u F = 1/x_u$.

Therefore, $a_u = 1$ and

$$g_u = \frac{x_u - up_u}{x_u^2} = \frac{u + (x_0 - u)e^{-t} - u(1 - e^{-t})}{x_u^2} = \frac{x_0 e^{-t}}{x_u^2} \geq 0. \quad (27a)$$

Hence

$$G_u = \int_0^t \frac{x_0 e^{-s}}{x_u(s)^2} ds \geq 0. \quad (27b)$$

Since $a_u = 1$ is constant, the kernel λ_u defined in Theorem 1 is

$$\lambda_u = \int_t^T e^{-(s-t)} ds = 1 - e^{-(T-t)}. \quad (27c)$$

In particular, $\lambda_u \geq 0$ for all $t \in [0, T]$, and $\dot{\lambda}_u = -e^{-(T-t)} < 0, \forall t \in [0, T]$.

Proposition 5 Consider the scalar System 3: $\dot{x}_u(t) = -x_u(t) + u$ and $\dot{y}_u = (u/x_u(t)) - y_u(t)$ with arbitrary $x_0 \geq 0$ and $y_0 = y_{ss} = 1$. Then, for every $T > 0$, the $\text{cDR}(u, T)$ is monotone nondecreasing with respect to u .

Proof 6 The sensitivity $q_u(t) = \partial_u y_u(t)$ satisfies $\dot{q}_u(t) + q_u(t) = g_u(t)$ where $g_u(t)$ is given in (27a). From (27c), $\lambda_u(t) \geq 0$ for all $t \in [0, T]$, and $g_u(t) \geq 0$ for all $t \geq 0$. Thus $\lambda_u(t)g_u(t) \geq 0, \forall t \in [0, T]$. By Theorem 1 (i), $\partial_u \text{cDR}(u, T) \geq 0$. Hence $u \mapsto \text{cDR}(u, T)$ is monotone nondecreasing.

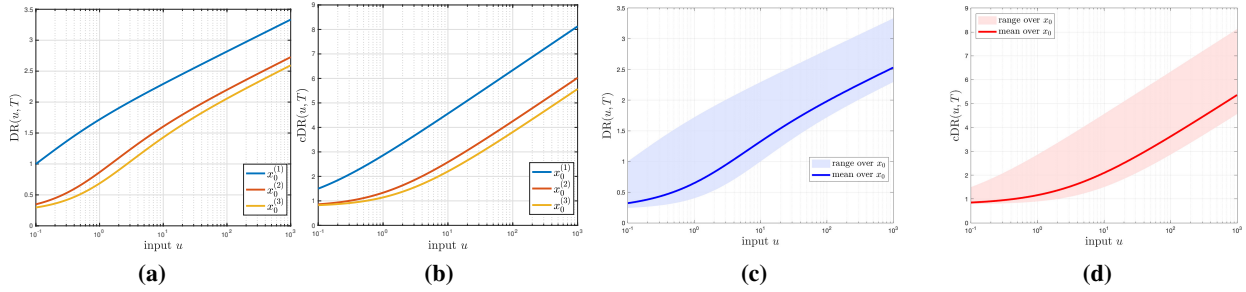


Figure 6: System 3: Dose response $\text{DR}(u, T)$ and cumulative dose response $\text{cDR}(u, T)$ for $u \in [10^{-1}, 10^3]$. (a)–(b) Responses for three fixed initial conditions x_0 . (c)–(d) Envelope (min–max band) and mean over $x_0 \in [0.1, 10]$. The results show that $\text{cDR}(u, T)$ is monotone nondecreasing with respect to u .

System 4: $\dot{y}_u = \frac{1}{x_u} - \frac{1}{u} y_u$. Here $F(x_u, y_u, u) = (1/x_u) - (y_u/u)$. Thus $\partial_x F = -1/x_u^2$, $\partial_y F = -1/u$, and $\partial_u F = y_u/u^2$. Therefore, $a_u = 1/u$ and $g_u = -(p_u/x_u^2) + (y_u/u^2)$. Hence

$$G_u = \int_0^t \left[-\frac{p_u(s)}{x_u(s)^2} + \frac{y_u(s)}{u^2} \right] ds. \quad (28a)$$

Since $a_u = 1/u$ is constant, the kernel λ_u defined in Theorem 1 is

$$\lambda_u = \int_t^T e^{-(s-t)/u} ds = u(1 - e^{-(T-t)/u}). \quad (28b)$$

In particular, $\lambda_u \geq 0$ for all $t \in [0, T]$, and $\dot{\lambda}_u = -e^{-(T-t)/u} < 0, \forall t \in [0, T]$. Using $x_u = up_u + x_0 e^{-t}$ and $\dot{y}_u = (1/x_u) - (y_u/u)$, we rewrite g_u as

$$g_u = -\frac{1}{u} \dot{y}_u + \frac{x_0 e^{-t}}{u x_u^2}. \quad (28c)$$

Integrating (28c) over $[0, t]$ and using $y_0 = 1$, we obtain

$$G_u = \frac{1 - y_u}{u} + \frac{x_0}{u} \int_0^t \frac{e^{-s}}{x_u(s)^2} ds. \quad (28d)$$

We now analyze the sign of $\partial_u \text{cDR}(u, T)$ for small and large values of u .

- First, assume that $0 < u \leq x_0$. Then $x_u = u + (x_0 - u)e^{-t} \geq u, \forall t \geq 0$. Evaluating the y -subsystem at $y_u = 1$ gives $\dot{y}_u|_{y_u=1} = (1/x_u) - (1/u) \leq 0$. Since $y_0 = 1$, the comparison principle yields $y_u \leq 1$ for all $t \geq 0$. Therefore, by (28d), $G_u \geq 0, \forall t \in [0, T]$. Moreover, $G_u \neq 0$ for $t > 0$ because the second term in (28d) is strictly positive. Since $-\dot{\lambda}_u > 0$, Theorem 1 (ii) implies $\partial_u \text{cDR}(u, T) > 0$ for all sufficiently small $u > 0$.

- Next, we consider the case of large u . Using variation of parameters formula on $\dot{y}_u + (1/u)y_u = 1/x_u$ and subtracting from $y_0 = 1$, we rewrite G_u in (28d) as

$$G_u = \frac{1 - e^{-t/u}}{u} - \frac{1}{u} \int_0^t \frac{e^{-(t-s)/u}}{x_u(s)} ds + \frac{x_0}{u} \int_0^t \frac{e^{-s}}{x_u(s)^2} ds. \quad (28e)$$

Since $x_u(s) = u + (x_0 - u)e^{-s}$, we have $x_u(s) \sim x_0 + us$ as $s \rightarrow 0$ (the dominant contribution comes from small s). It follows that, for each fixed $t > 0$,

$$\int_0^t \frac{e^{-(t-s)/u}}{x_u(s)} ds \sim \int_0^t \frac{1}{x_0 + us} ds = \frac{1}{u} \log \left(1 + \frac{ut}{x_0} \right) \sim \frac{\log u}{u}, \quad u \rightarrow \infty.$$

On the other hand,

$$\frac{1 - e^{-t/u}}{u} = O(u^{-2}), \quad \frac{x_0}{u} \int_0^t \frac{e^{-s}}{x_u(s)^2} ds = O(u^{-2}).$$

Therefore,

$$G_u = -\frac{\log u}{u^2} + O(u^{-2}), \quad u \rightarrow \infty. \quad (28f)$$

In particular, for every fixed $t > 0$, one has $G_u < 0$ for all sufficiently large u .

Proposition 6 Consider the scalar System 4: $\dot{x}_u(t) = -x_u(t) + u$ and $\dot{y}_u(t) = (1/x_u(t)) - (y_u(t)/u)$ with arbitrary $x_0 > 0$ and $y_0 = y_{ss} = 1$. Then there exist $u_-, u_+ > 0$ such that

$$\partial_u \text{cDR}(u_-, T) > 0, \quad \partial_u \text{cDR}(u_+, T) < 0.$$

In particular, $\text{cDR}(u, T)$ is not monotone with respect to u .

Proof 7 The sensitivity $q_u(t) = \partial_u y_u(t)$ satisfies $\dot{q}_u(t) + (1/u)q_u(t) = g_u(t)$, where $g_u(t)$ is given by (28c). Also, $-\dot{\lambda}_u(t) > 0$ for all $t \in [0, T]$.

1. If $0 < u \leq x_0$, then $x_u(t) \geq u$ and thus $y_u(t) \leq 1$ for all $t \geq 0$. By (28d), this implies $G_u(t) \geq 0$ on $[0, T]$, with $G_u \not\equiv 0$. Hence, by Theorem 1 (ii), $\partial_u \text{cDR}(u, T) > 0$.
2. For large u , by (28f), one has $G_u(t) < 0$ for every fixed $t > 0$ and all sufficiently large u . Therefore, $\partial_u \text{cDR}(u, T) < 0$.

Thus, $\partial_u \text{cDR}(u, T) > 0$ for small u and $\partial_u \text{cDR}(u, T) < 0$ for large u , and hence $\text{cDR}(u, T)$ is not monotone.

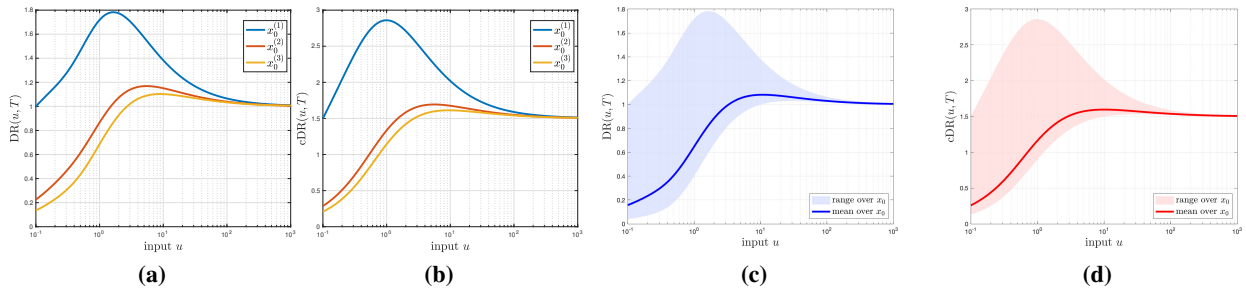


Figure 7: System 4: Dose response $\text{DR}(u, T)$ and cumulative dose response $\text{cDR}(u, T)$ for $u \in [10^{-1}, 10^3]$. (a)–(b) Responses for three fixed initial conditions x_0 . (c)–(d) Envelope (min–max band) and mean over $x_0 \in [0.1, 10]$. The results show that $\text{cDR}(u, T)$ is not monotone with respect to u , exhibiting a change in monotonicity.

System 5: $\dot{y}_u = u - x_u y_u$. Here $F(x_u, y_u, u) = u - x_u y_u$. Thus $\partial_x F = -y_u$, $\partial_y F = -x_u$, and $\partial_u F = 1$.

Therefore, $a_u = x_u$ and $g_u = 1 - y_u p_u$. Hence

$$G_u = \int_0^t [1 - y_u(s)p_u(s)] ds = t - \int_0^t y_u(s)p_u(s) ds. \quad (29a)$$

Since $a_u = x_u \geq 0$, the kernel λ_u defined in Theorem 1 satisfies

$$\lambda_u = \int_t^T \exp\left(-\int_t^s x_u(\tau) d\tau\right) ds. \quad (29b)$$

In particular, $\lambda_u \geq 0$ for all $t \in [0, T]$.

We now distinguish two cases.

- If $x_0 \geq u$, then $x_u = u + (x_0 - u)e^{-t} \geq u, \forall t \geq 0$. Since $y_0 = y_{ss} = 1$, evaluating the y -subsystem at $y_u = 1$ gives $\dot{y}_u|_{y_u=1} = u - x_u \leq 0$. Hence, by comparison principle, $y_u \leq 1$ for all $t \geq 0$. Moreover, since $0 \leq p_u := 1 - e^{-t} \leq 1$, it follows that $y_u p_u \leq 1$, and therefore

$$g_u = 1 - y_u p_u \geq 0. \quad (29c)$$

- Next, suppose that $x_0 < u$. Then $x_u = u + (x_0 - u)e^{-t} \leq u, \forall t \geq 0$. Again, since $y_0 = 1$, evaluating at $y_u = 1$ gives $\dot{y}_u|_{y_u=1} = u - x_u \geq 0$, so by comparison principle $y_u \geq 1$ for all $t \geq 0$. Using $x_u = u + (x_0 - u)e^{-t}$ in the y -subsystem, we rewrite g_u as

$$\begin{aligned} g_u &:= 1 - y_u p_u = 1 - (1 - e^{-t})y_u \\ &= 1 - y_u + \frac{u - x_0 + x_0}{u} e^{-t} y_u \\ &= 1 - \frac{x_u}{u} y_u + \frac{x_0}{u} e^{-t} y_u =: \frac{1}{u} \dot{y}_u + \frac{x_0}{u} e^{-t} y_u. \end{aligned} \quad (29d)$$

Integrating (29d) over $[0, t]$ and using $y_0 = 1$, we obtain

$$G_u = \frac{1}{u} (y_u - 1) + \frac{x_0}{u} \int_0^t e^{-s} y_u(s) ds \geq 0, \quad (29e)$$

since $y_u \geq 1$ and $y_u \geq 0$. Thus, both terms in (29e) are nonnegative and hence $G_u \geq 0$.

It remains to show that $\dot{\lambda}_u \leq 0$ when $x_0 < u$. Differentiating (29b) gives $\dot{\lambda}_u = -1 + x_u \lambda_u$. Since $x_0 < u$, the function $x_u = u + (x_0 - u)e^{-t}$ is nondecreasing. Thus, for every $s \in [t, T]$, $x_u(\tau) \geq x_u, \forall \tau \in [t, s]$. Therefore,

$$\exp\left(-\int_t^s x_u(\tau) d\tau\right) \leq e^{-(s-t)x_u}.$$

Integrating from t to T yields

$$x_u \lambda_u \leq x_u \int_t^T e^{-(s-t)x_u} ds = 1 - e^{-(T-t)x_u} \leq 1.$$

Hence

$$\dot{\lambda}_u = -1 + x_u \lambda_u \leq 0. \quad (29f)$$

Proposition 7 Consider the scalar System 5: $\dot{x}_u(t) = -x_u(t) + u$ and $\dot{y}_u(t) = u - x_u(t)y_u(t)$ with arbitrary $x_0 \geq 0$ and $y_0 = y_{ss} = 1$. Then, for every $T > 0$, the $\text{cDR}(u, T)$ is monotone nondecreasing with respect to u .

Proof 8 The sensitivity $q_u(t) = \partial_u y_u(t)$ satisfies $\dot{q}_u(t) + x_u(t)q_u(t) = g_u(t)$ where $g_u(t)$ is given in (29d).

1. If $x_0 \geq u$, then by (29c), $g_u(t) \geq 0$ for all $t \in [0, T]$. Since $\lambda_u(t) \geq 0$ by (29b), it follows that $\lambda_u(t)g_u(t) \geq 0, \forall t \in [0, T]$. Hence, by Theorem 1 (i), $\partial_u \text{cDR}(u, T) \geq 0$.
2. If $x_0 < u$, then by (29e), $G_u(t) \geq 0$ for all $t \in [0, T]$, and by (29f), $\dot{\lambda}_u(t) \leq 0$ for all $t \in [0, T]$. Thus $-\dot{\lambda}_u(t)G_u(t) \geq 0, \forall t \in [0, T]$. Hence, by Theorem 1 (ii), $\partial_u \text{cDR}(u, T) \geq 0$.

Therefore, in either case, $u \mapsto \text{cDR}(u, T)$ is monotone nondecreasing.

Monotonicity of cumulative dose response $cDR(u, T)$

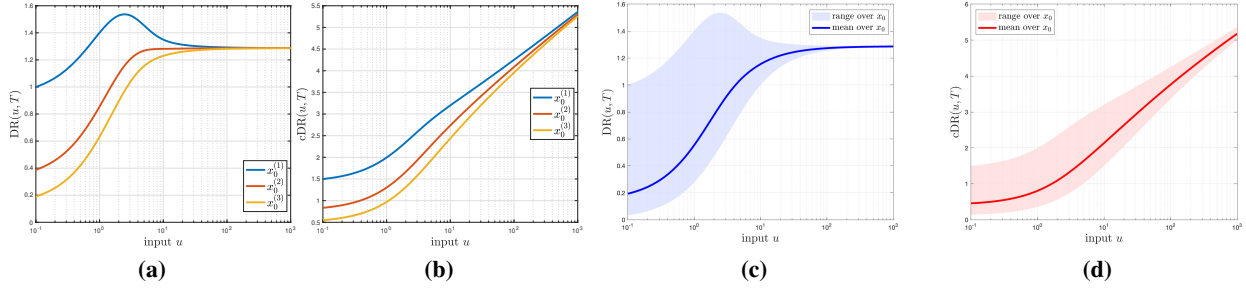


Figure 8: System 5: Dose response $DR(u, T)$ and cumulative dose response $cDR(u, T)$ for $u \in [10^{-1}, 10^3]$. (a)–(b) Responses for three fixed initial conditions x_0 . (c)–(d) Envelope (min–max band) and mean over $x_0 \in [0.1, 10]$. The results show that $cDR(u, T)$ is monotone nondecreasing with respect to u .

System 6: $\dot{y}_u = 1 - \frac{x_u}{u}y_u$. Here $F(x_u, y_u, u) = 1 - (x_u y_u / u)$. Thus $\partial_x F = -y_u / u$, $\partial_y F = -x_u / u$, and $\partial_u F = x_u y_u / u^2$. Therefore, $a_u = x_u / u$ and

$$g_u = \frac{y_u(x_u - up_u)}{u^2} = \frac{x_0 e^{-t}}{u^2} y_u \geq 0. \quad (30a)$$

Hence

$$G_u = \int_0^t \frac{x_0 e^{-s}}{u^2} y_u(s) ds \geq 0. \quad (30b)$$

Since $a_u = x_u / u \geq 0$, the kernel λ_u defined in Theorem 1 satisfies

$$\lambda_u = \int_t^T \exp\left(-\int_t^s \frac{x_u(\tau)}{u} d\tau\right) ds. \quad (30c)$$

In particular, $\lambda_u \geq 0$ for all $t \in [0, T]$.

Proposition 8 Consider the scalar System 6: $\dot{x}_u(t) = -x_u(t) + u$ and $\dot{y}_u(t) = 1 - (x_u(t)y_u(t)/u)$ with arbitrary $x_0 \geq 0$ and $y_0 = y_{ss} = 1$. Then, for every $T > 0$, the $cDR(u, T)$ is monotone nondecreasing with respect to u .

Proof 9 The sensitivity $q_u(t) = \partial_{x_u} y_u(t)$ satisfies $\dot{q}_u(t) + (x_u(t)/u)q_u(t) = g_u(t)$ where $g_u(t)$ is given in (30a). From (30c), $\lambda_u(t) \geq 0$ for all $t \in [0, T]$, and $g_u(t) \geq 0$ for all $t \geq 0$. Thus $\lambda_u(t)g_u(t) \geq 0, \forall t \in [0, T]$. By Theorem 1 (i), $\partial_u cDR(u, T) \geq 0$. Hence $u \mapsto cDR(u, T)$ is monotone nondecreasing.

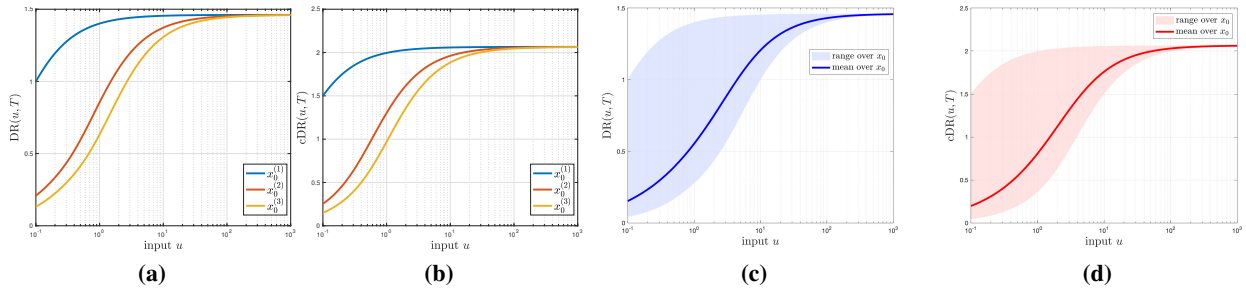


Figure 9: System 6: Dose response $DR(u, T)$ and cumulative dose response $cDR(u, T)$ for $u \in [10^{-1}, 10^3]$. (a)–(b) Responses for three fixed initial conditions x_0 . (c)–(d) Envelope (min–max band) and mean over $x_0 \in [0.1, 10]$. The results show that $cDR(u, T)$ is monotone nondecreasing with respect to u .

System 7: $\dot{y}_u = \frac{1}{u} - \frac{1}{x_u}y_u$. Here $F(x_u, y_u, u) = (1/u) - (y_u/x_u)$. Thus $\partial_x F = y_u/x_u^2$, $\partial_y F = -1/x_u$, and $\partial_u F = -1/u^2$. Therefore, $a_u = 1/x_u$ and $g_u = (y_u p_u/x_u^2) - (1/u^2)$. Using $x_u = up_u + x_0 e^{-t}$ and $\dot{y}_u = (1/u) - (y_u/x_u)$,

we rewrite g_u as

$$g_u = -\frac{1}{u} \dot{y}_u - \frac{x_0 e^{-t}}{u} \frac{y_u}{x_u^2}. \quad (31a)$$

Integrating (31a) over $[0, t]$ and using $y_0 = 1$, we obtain

$$G_u = \frac{1 - y_u}{u} - \frac{x_0}{u} \int_0^t \frac{e^{-s} y_u(s)}{x_u(s)^2} ds. \quad (31b)$$

Since $a_u = 1/x_u > 0$, the kernel λ_u defined in Theorem 1 satisfies

$$\lambda_u = \int_t^T \exp\left(-\int_t^s \frac{1}{x_u(\tau)} d\tau\right) ds. \quad (31c)$$

In particular, $\lambda_u \geq 0$ for all $t \in [0, T]$.

We now analyze two cases.

- If $0 < u \leq x_0$, then $x_u = u + (x_0 - u)e^{-t} \geq u$ for all $t \geq 0$. Evaluating the y -subsystem at $y_u = 1$ gives $\dot{y}_u|_{y_u=1} = (1/u) - (1/x_u) \geq 0$. Since $y_u(0) = 1$, the comparison principle yields $y_u \geq 1$ for all $t \geq 0$. By (31b), this implies $G_u \leq 0, \forall t \in [0, T]$. Moreover, since x_u is nonincreasing in this case, one has $\dot{\lambda}_u \leq 0$ exactly as in the proof of System 5. Hence $-\dot{\lambda}_u G_u \leq 0, \forall t \in [0, T]$, and therefore, by Theorem 1 (ii), $\partial_u \text{cDR}(u, T) < 0$.
- Next, we consider the case of large u . Let $z_u := 1 - y_u$. Then

$$\dot{z}_u + \frac{1}{x_u} z_u = \frac{1}{x_u} - \frac{1}{u} = \frac{u - x_0}{u} \frac{e^{-t}}{x_u}, \quad z_u(0) = 0.$$

By variation of parameters,

$$z_u = \frac{u - x_0}{u} \int_0^t \exp\left(-\int_s^t \frac{1}{x_u(r)} dr\right) \frac{e^{-s}}{x_u(s)} ds.$$

Since $x_u(s) = u + (x_0 - u)e^{-s} \sim x_0 + us$ as $s \rightarrow 0$, it follows that, for each fixed $t > 0$,

$$z_u \sim \frac{\log u}{u}, \quad u \rightarrow \infty. \quad (31d)$$

On the other hand, the second term in (31b) satisfies

$$\frac{x_0}{u} \int_0^t \frac{e^{-s} y_u(s)}{x_u(s)^2} ds = O(u^{-2}). \quad (31e)$$

Substituting (31d) and (31e) into (31b), we obtain

$$G_u = \frac{\log u}{u^2} + O(u^{-2}), \quad u \rightarrow \infty. \quad (31f)$$

In particular, for every fixed $t > 0$, one has $G_u > 0$ for all sufficiently large u .

Proposition 9 Consider the scalar System 7 $\dot{x}_u(t) = -x_u(t) + u$ and $\dot{y}_u(t) = \frac{1}{u} - \frac{1}{x_u(t)} y_u(t)$ with arbitrary $x_0 > 0$ and $y_0 = y_{ss} = 1$. Then there exist $u_-, u_+ > 0$ such that

$$\partial_u \text{cDR}(u_-, T) < 0, \quad \partial_u \text{cDR}(u_+, T) > 0.$$

In particular, $\text{cDR}(u, T)$ is not monotone with respect to u .

Proof 10 The sensitivity $q_u(t) = \partial_u y_u(t)$ satisfies $\dot{q}_u(t) + \frac{1}{x_u(t)} q_u(t) = g_u(t)$, where $g_u(t)$ is given by (31a).

1. If $0 < u \leq x_0$, then $x_u(t) \geq u$ and therefore $y_u(t) \geq 1$ for all $t \geq 0$. By (31b), this yields $G_u(t) \leq 0$ for all $t \in [0, T]$. Moreover, since $\dot{\lambda}_u(t) \leq 0$, Theorem 1 (ii) implies $\partial_u \text{cDR}(u, T) < 0$.
2. For large u , (31f) shows that, for every fixed $t > 0$, one has $G_u(t) > 0$ for all sufficiently large u . Arguing exactly as in the large- u part of System 4, this yields $\partial_u \text{cDR}(u, T) > 0$ for all sufficiently large u .

Thus there exist $u_-, u_+ > 0$ such that $\partial_u \text{cDR}(u_-, T) < 0$ and $\partial_u \text{cDR}(u_+, T) > 0$, and therefore $\text{cDR}(u, T)$ is not monotone.

Monotonicity of cumulative dose response $cDR(u, T)$

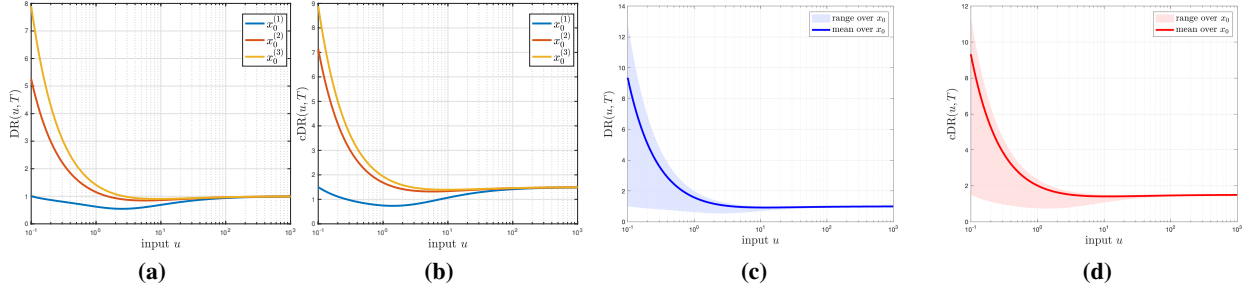


Figure 10: System 7: Dose response $DR(u, T)$ and cumulative dose response $cDR(u, T)$ for $u \in [10^{-1}, 10^3]$. (a)–(b) Responses for three fixed initial conditions x_0 . (c)–(d) Envelope (min–max band) and mean over $x_0 \in [0.1, 10]$. The results show that $cDR(u, T)$ is not monotone with respect to u , exhibiting a change in monotonicity.

System 8: $\dot{y}_u = 1 - \frac{u}{x_u} y_u$. Here $F(x_u, y_u, u) = 1 - (u y_u / x_u)$. Thus $\partial_x F = u y_u / x_u^2$, $\partial_y F = -u / x_u$, and $\partial_u F = -y_u / x_u$. Therefore, $a_u = u / x_u$ and

$$g_u = \frac{u y_u}{x_u^2} p_u - \frac{y_u}{x_u} = \frac{y_u (u p_u - x_u)}{x_u^2} = -\frac{x_0 e^{-t}}{x_u^2} y_u \leq 0. \quad (32a)$$

Hence

$$G_u = -\int_0^t \frac{x_0 e^{-s}}{x_u(s)^2} y_u(s) ds \leq 0. \quad (32b)$$

Since $a_u = u / x_u \geq 0$, the kernel λ_u defined in Theorem 1 satisfies

$$\lambda_u = \int_t^T \exp\left(-\int_t^s \frac{u}{x_u(\tau)} d\tau\right) ds. \quad (32c)$$

In particular, $\lambda_u \geq 0$ for all $t \in [0, T]$.

Proposition 10 Consider the scalar System 8: $\dot{x}_u(t) = -x_u(t) + u$ and $\dot{y}_u(t) = 1 - (u y_u(t)) / x_u(t)$ with arbitrary $x_0 \geq 0$ and $y_0 = y_{ss} = 1$. Then, for every $T > 0$, the $cDR(u, T)$ is monotone nonincreasing with respect to u .

Proof 11 The sensitivity $q_u(t) = \partial_u y_u(t)$ satisfies $\dot{q}_u(t) + (u / x_u(t)) q_u(t) = g_u(t)$ where $g_u(t)$ is given in (32a). From (32c), $\lambda_u(t) \geq 0$ for all $t \in [0, T]$, and $g_u(t) \leq 0$ for all $t \geq 0$. Thus $\lambda_u(t) g_u(t) \leq 0, \forall t \in [0, T]$. By Theorem 1 (i), $\partial_u cDR(u, T) \leq 0$. Hence $u \mapsto cDR(u, T)$ is monotone nonincreasing.

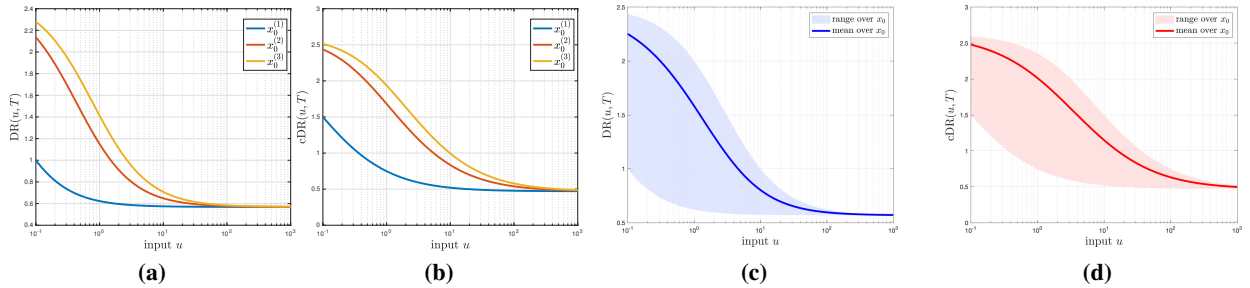


Figure 11: System 8: Dose response $DR(u, T)$ and cumulative dose response $cDR(u, T)$ for $u \in [10^{-1}, 10^3]$. (a)–(b) Responses for three fixed initial conditions x_0 . (c)–(d) Envelope (min–max band) and mean over $x_0 \in [0.1, 10]$. The results show that $cDR(u, T)$ is monotone nonincreasing with respect to u .

C Monotonicity of $\text{cDR}(u, T)$ for the generalized IFFM systems

In this subsection, we compute the quantities $a_u(t)$, $g_u(t)$, and

$$G_u(t) := \int_0^t g_u(s) ds$$

for the scalar IFFM systems.

For all systems, we consider $\dot{x}_u = Ax_u + bu$, with initial condition $x_u(0) = x_0$, independent of u . Differentiating the x -subsystem with respect to u gives $\dot{p}_u = Ap_u + b$ with $p_u(0) = 0$ where $p_u := \partial_u x_u \in \mathbb{R}^n$. Hence,

$$p_u(t) = \int_0^t e^{A(t-s)} b ds = A^{-1}(e^{At} - I)b. \quad (33)$$

Since A is Metzler, $e^{At} \geq 0$ for all $t \geq 0$, and since $b \geq 0$, it follows that $p_u(t) \geq 0, \forall t \geq 0$. Moreover, by variation of parameters formula,

$$x_u(t) = e^{At}x_0 + \int_0^t e^{A(t-s)}bu ds = e^{At}x_0 + up_u(t). \quad (34)$$

Let $q_u := \partial_u y_u$. If $\dot{y}_u = F(x_u, y_u, u)$, then differentiating with respect to u yields

$$\dot{q}_u = \partial_x F(x_u, y_u, u) p_u + \partial_y F(x_u, y_u, u) q_u + \partial_u F(x_u, y_u, u). \quad (35)$$

Rewriting (35) in the form $\dot{q}_u + a_u(t)q_u = g_u(t)$, we identify

$$a_u(t) = -\partial_y F(x_u, y_u, u), \quad g_u(t) = \partial_x F(x_u, y_u, u) p_u + \partial_u F(x_u, y_u, u).$$

Vector case of System 1

We consider the vector analogue of System 1:

$$\dot{x}_u = Ax_u + bu, \quad \dot{y}_u = \frac{c^\top x_u}{\beta u} - dy_u, \quad (36a)$$

where $A \in \mathbb{R}^{n \times n}$ is Metzler and Hurwitz, $b, c \in \mathbb{R}_+^n$, and $\beta, d \in \mathbb{R}_+$. We assume that the initial condition $x_u(0) = x_0$ is independent of u , with $x_0 \geq 0$, and that $y_u(0) = y_0$ is also independent of u .

Let $q_u := \partial_u y_u \in \mathbb{R}$. Differentiating the y -equation in (36a) with respect to u gives

$$\dot{q}_u = \frac{c^\top p_u}{\beta u} - \frac{c^\top x_u}{\beta u^2} - dq_u, \quad q_u(0) = 0.$$

Therefore, q_u satisfies $\dot{q}_u + a_u q_u = g_u$, $q_u(0) = 0$ with

$$a_u = d, \quad g_u = \frac{c^\top p_u}{\beta u} - \frac{c^\top x_u}{\beta u^2}.$$

Using (34), we rewrite g_u as

$$g_u = \frac{uc^\top p_u - c^\top x_u}{\beta u^2} = \frac{uc^\top p_u - c^\top [e^{At}x_0 + up_u]}{\beta u^2} = -\frac{c^\top e^{At}x_0}{\beta u^2}. \quad (36b)$$

Since $e^{At} \geq 0$, $c \in \mathbb{R}_+^n$, and $x_0 \geq 0$, we conclude that $g_u(t) \leq 0, \forall t \geq 0$. Integrating g_u over $[0, t]$, we obtain

$$G_u := \int_0^t g_u(s) ds = -\frac{1}{\beta u^2} \int_0^t c^\top e^{As} x_0 ds \leq 0. \quad (36c)$$

Since $a_u = d$ is constant, the kernel λ_u defined in Theorem 1 is

$$\lambda_u = \int_t^T e^{-d(s-t)} ds = \frac{1 - e^{-d(T-t)}}{d}. \quad (36d)$$

Therefore, $\lambda_u \geq 0, \forall t \in [0, T]$.

Proposition 11 Consider the system (36a), where A is Metzler and Hurwitz, $b, c \in \mathbb{R}_+^n$, $\beta, d \in \mathbb{R}_+$, and $x_0 \geq 0$ is independent of u . Then, for every $T > 0$, the $\text{cDR}(u, T)$ is monotone nonincreasing with respect to u .

Proof 12 The sensitivity $q_u = \partial_u y_u$ satisfies $\dot{q}_u + dq_u = g_u$ in which g_u is given in (36b). Also, $\lambda_u \geq 0, \forall t \in [0, T]$ by (36d). Thus $\lambda_u g_u \leq 0, \forall t \in [0, T]$. By Theorem 1 (i), $\partial_u \text{cDR}(u, T) \leq 0$. Hence $u \mapsto \text{cDR}(u, T)$ is monotone nonincreasing.

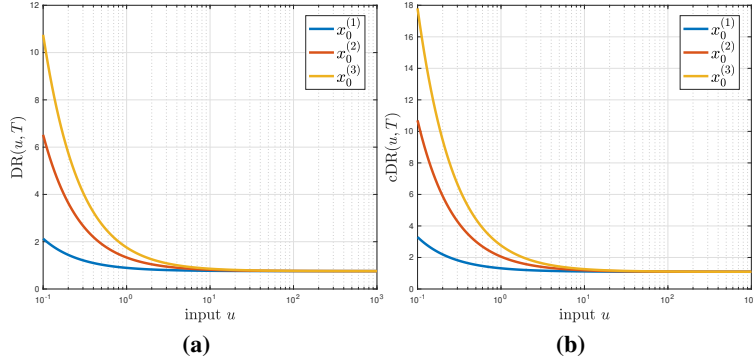


Figure 12: Vector System 1: Dose response $\text{DR}(u, T)$ and cumulative dose response $\text{cDR}(u, T)$ for $u \in [10^{-1}, 10^3]$. Results are shown for three fixed initial conditions x_0 . The curves illustrate that $\text{cDR}(u, T)$ is monotone nonincreasing with respect to u .

Vector case of System 3

We consider the vector analogue of System 3:

$$\dot{x}_u = Ax_u + bu, \quad \dot{y}_u = \frac{\beta u}{K + c^\top x_u} - dy_u, \quad (37a)$$

where $A \in \mathbb{R}^{n \times n}$ is Metzler and Hurwitz, $b, c \in \mathbb{R}_+^n$, and $K, \beta, d \in \mathbb{R}_+$. We assume that the initial condition $x_u(0) = x_0$ is independent of u , with $x_0 \geq 0$, and that $y_u(0) = y_0$ is also independent of u .

Let $q_u := \partial_u y_u \in \mathbb{R}$. Differentiating the y -equation in (37a) with respect to u gives

$$\dot{q}_u = \frac{\beta}{K + c^\top x_u} - \frac{\beta u c^\top p_u}{(K + c^\top x_u)^2} - dq_u, \quad q_u(0) = 0.$$

Therefore, q_u satisfies $\dot{q}_u + a_u q_u = g_u$, $q_u(0) = 0$ with

$$a_u = d, \quad g_u = \frac{\beta}{K + c^\top x_u} - \frac{\beta u c^\top p_u}{(K + c^\top x_u)^2}.$$

Using (34), we rewrite g_u as

$$g_u = \frac{\beta [K + c^\top x_u - u c^\top p_u]}{(K + c^\top x_u)^2} = \frac{\beta [K + c^\top e^{At} x_0]}{(K + c^\top x_u)^2}. \quad (37b)$$

Since $K > 0$, $e^{At} \geq 0$, $c \in \mathbb{R}_+^n$, and $x_0 \geq 0$, we conclude that $g_u \geq 0, \forall t \geq 0$. Integrating g_u over $[0, t]$, we obtain

$$G_u := \int_0^t g_u(s) ds \geq 0. \quad (37c)$$

Since $a_u = d$ is constant, the kernel λ_u defined in Theorem 1 is

$$\lambda_u = \int_t^T e^{-d(s-t)} ds = \frac{1 - e^{-d(T-t)}}{d}. \quad (37d)$$

Therefore, $\lambda_u \geq 0, \forall t \in [0, T]$.

Proposition 12 Consider the system (37a), where A is Metzler and Hurwitz, $b, c \in \mathbb{R}_+^n$, $K, \beta, d \in \mathbb{R}_+$, and $x_0 \geq 0$ is independent of u . Then, for every $T > 0$, the $\text{cDR}(u, T)$ is monotone nondecreasing with respect to u .

Proof 13 The sensitivity $q_u(t) = \partial_u y_u(t)$ satisfies $\dot{q}_u(t) + dq_u(t) = g_u(t)$ in which $g_u(t)$ is given in (37b). Also, $\lambda_u(t) \geq 0, \forall t \in [0, T]$ by (37d). Thus $\lambda_u(t)g_u(t) \geq 0, \forall t \in [0, T]$. By Theorem 1 (i), $\partial_u \text{cDR}(u, T) \geq 0$. Hence $u \mapsto \text{cDR}(u, T)$ is monotone nondecreasing.

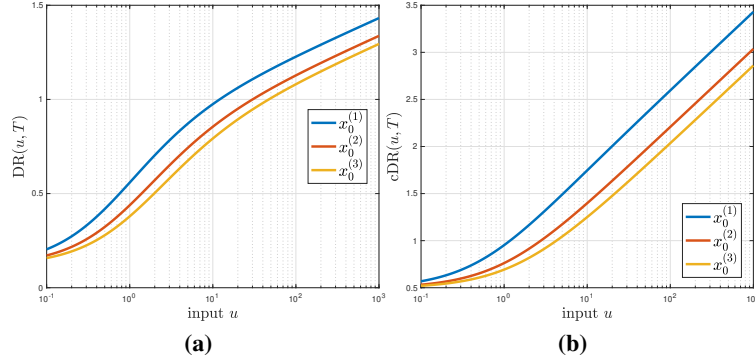


Figure 13: Vector System 3: Dose response $\text{DR}(u, T)$ and cumulative dose response $\text{cDR}(u, T)$ for $u \in [10^{-1}, 10^3]$. Results are shown for three fixed initial conditions x_0 . The curves illustrate that $\text{cDR}(u, T)$ is monotone nondecreasing with respect to u .

Vector case of System 6

We consider the vector analogue of System 6:

$$\dot{x}_u = Ax_u + bu, \quad \dot{y}_u = d - \frac{c^\top x_u}{\beta u} y_u, \quad (38a)$$

where $A \in \mathbb{R}^{n \times n}$ is Metzler and Hurwitz, $b, c \in \mathbb{R}_+^n$, and $\beta, d \in \mathbb{R}_+$. We assume that the initial condition $x_u(0) = x_0$ is independent of u , with $x_0 \geq 0$, and that $y_u(0) = y_0$ is also independent of u .

Let $q_u := \partial_u y_u \in \mathbb{R}$. Differentiating the y -equation in (38a) with respect to u gives

$$\dot{q}_u = -\frac{c^\top p_u}{\beta u} y_u + \frac{c^\top x_u}{\beta u^2} y_u - \frac{c^\top x_u}{\beta u} q_u, \quad q_u(0) = 0.$$

Therefore, q_u satisfies $\dot{q}_u + a_u q_u = g_u$, $q_u(0) = 0$ with

$$a_u = \frac{c^\top x_u}{\beta u}, \quad g_u = -\frac{c^\top p_u}{\beta u} y_u + \frac{c^\top x_u}{\beta u^2} y_u.$$

Using (34), we rewrite g_u as

$$g_u = \frac{(c^\top x_u - u c^\top p_u) y_u}{\beta u^2} = \frac{c^\top e^{At} x_0}{\beta u^2} y_u. \quad (38b)$$

Since $e^{At} \geq 0$, $c \in \mathbb{R}_+^n$, $x_0 \geq 0$, and $y_u \geq 0$, we conclude that $g_u \geq 0$, $\forall t \geq 0$. Integrating g_u over $[0, t]$, we obtain

$$G_u := \int_0^t g_u(s) ds = \frac{1}{\beta u^2} \int_0^t c^\top e^{As} x_0 y_u(s) ds \geq 0. \quad (38c)$$

Since $a_u = (c^\top x_u)/(\beta u) \geq 0$, the kernel λ_u defined in Theorem 1 satisfies

$$\lambda_u = \int_t^T \exp\left(-\int_t^s \frac{c^\top x_u(\tau)}{\beta u} d\tau\right) ds. \quad (38d)$$

Therefore, $\lambda_u \geq 0$, $\forall t \in [0, T]$.

Proposition 13 Consider the system (38a), where A is Metzler and Hurwitz, $b, c \in \mathbb{R}_+^n$, $\beta, d \in \mathbb{R}_+$, and $x_0 \geq 0$ is independent of u . Then, for every $T > 0$, the $\text{cDR}(u, T)$ is monotone nondecreasing with respect to u .

Proof 14 The sensitivity $q_u(t) = \partial_u y_u(t)$ satisfies $\dot{q}_u(t) + \frac{c^\top x_u(t)}{\beta u} q_u(t) = g_u(t)$ in which $g_u(t)$ is given in (38b). Also, $\lambda_u(t) \geq 0$, $\forall t \in [0, T]$ by (38d). Thus $\lambda_u(t) g_u(t) \geq 0$, $\forall t \in [0, T]$. By Theorem 1 (i), $\partial_u \text{cDR}(u, T) \geq 0$. Hence $u \mapsto \text{cDR}(u, T)$ is monotone nondecreasing.

Monotonicity of cumulative dose response $\text{cDR}(u, T)$

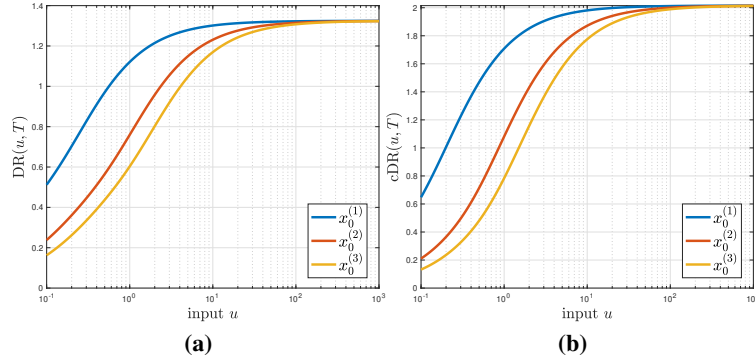


Figure 14: Vector System 6: Dose response $\text{DR}(u, T)$ and cumulative dose response $\text{cDR}(u, T)$ for $u \in [10^{-1}, 10^3]$. Results are shown for three fixed initial conditions x_0 . The curves illustrate that $\text{cDR}(u, T)$ is monotone nondecreasing with respect to u .

Vector case of System 7

We consider the vector analogue of System 7:

$$\dot{x}_u = Ax_u + bu, \quad \dot{y}_u = \frac{1}{\beta u} - \frac{dy_u}{K + c^\top x_u}, \quad (39a)$$

where $A \in \mathbb{R}^{n \times n}$ is Metzler and Hurwitz, $b, c \in \mathbb{R}_+^n$, and $K, \beta, d \in \mathbb{R}_+$. We assume that the initial condition $x_u(0) = x_0$ is independent of u , with $x_0 \geq 0$, and that $y_u(0) = y_0$ is also independent of u .

Let $q_u := \partial_u y_u \in \mathbb{R}$. Differentiating the y -equation in (39a) with respect to u gives

$$\dot{q}_u = -\frac{1}{\beta u^2} + \frac{dc^\top p_u}{(K + c^\top x_u)^2} y_u - \frac{d}{K + c^\top x_u} q_u, \quad q_u(0) = 0.$$

Therefore, q_u satisfies $\dot{q}_u + a_u q_u = g_u$, $q_u(0) = 0$ with

$$a_u = \frac{d}{K + c^\top x_u}, \quad g_u = \frac{dc^\top p_u}{(K + c^\top x_u)^2} y_u - \frac{1}{\beta u^2}. \quad (39b)$$

Integrating g_u over $[0, t]$, we obtain

$$G_u := \int_0^t g_u(s) ds = \int_0^t \left[\frac{dc^\top p_u(s)}{(K + c^\top x_u(s))^2} y_u(s) - \frac{1}{\beta u^2} \right] ds. \quad (39c)$$

Since $a_u = d/(K + c^\top x_u) \geq 0$, the kernel λ_u defined in Theorem 1 satisfies

$$\lambda_u = \int_t^T \exp\left(-\int_t^s \frac{d}{K + c^\top x_u(\tau)} d\tau\right) ds. \quad (39d)$$

Therefore, $\lambda_u \geq 0, \forall t \in [0, T]$.

Proposition 14 Consider the system (39a), where A is Metzler and Hurwitz, $b, c \in \mathbb{R}_+^n$, $K, \beta, d \in \mathbb{R}_+$, and $x_0 \geq 0$ is independent of u . Then there exist $u_-, u_+ > 0$ such that

$$\partial_u \text{cDR}(u_-, T) < 0, \quad \partial_u \text{cDR}(u_+, T) > 0.$$

In particular, $\text{cDR}(u, T)$ is not monotone with respect to u .

Proof 15 The sensitivity $q_u(t) = \partial_u y_u(t)$ satisfies $\dot{q}_u(t) + a_u(t)q_u(t) = g_u(t)$, where $g_u(t)$ is given in (39c).

For small u , let $M_p := \max_{t \in [0, T]} c^\top p_u(t) < \infty$. Also, since

$$\dot{y}_u(t) = \frac{1}{\beta u} - \frac{dy_u(t)}{K + c^\top x_u(t)} \leq \frac{1}{\beta u},$$

it follows that $y_u(t) \leq y_0 + (T/\beta u)$, $\forall t \in [0, T]$. Because $K + c^\top x_u(t) \geq K$, we obtain

$$g_u(t) \leq \frac{dM_p}{K^2} \left(y_0 + \frac{T}{\beta u} \right) - \frac{1}{\beta u^2}, \quad \forall t \in [0, T].$$

Hence $g_u(t) < 0$ for all $t \in [0, T]$ and all sufficiently small $u > 0$. Since $\lambda_u(t) \geq 0$ by (39d), Theorem 1 (i) implies $\partial_u \text{cDR}(u, T) < 0$ for all sufficiently small $u > 0$.

Case: small u . If $u < K/(\beta dy_0)$, then for every $t \in [0, T]$,

$$\dot{y}_u|_{y_u=y_0} = \frac{1}{\beta u} - \frac{dy_0}{K + c^\top x_u(t)} \geq \frac{1}{\beta u} - \frac{dy_0}{K} > 0.$$

Since $y_u(0) = y_0$, the comparison principle yields $y_u \geq y_0$, $t \in [0, T]$. Next, using $x_u = e^{At}x_0 + up_u$ and $\dot{y}_u$ in (39a) we rewrite g_u as

$$g_u = -\frac{1}{u}\dot{y}_u - \frac{d(K + c^\top e^{At}x_0)}{u(K + c^\top x_u)^2}y_u. \quad (40)$$

Integrating (40) over $[0, t]$, we obtain

$$G_u = \frac{y_0 - y_u}{u} - \frac{d}{u} \int_0^t \frac{[K + c^\top e^{As}x_0]y_u(s)}{(K + c^\top x_u(s))^2} ds. \quad (41)$$

Since $y_u \geq y_0$ and the second term in (41) is nonpositive, it follows that $G_u \leq 0$, $t \in [0, T]$. Moreover, $G_u \neq 0$ because the second term in (41) is strictly negative.

Next, we consider the case of large u . By variation of parameters formula,

$$y_u(t) = y_0 e^{-h_u(t)} + \frac{1}{\beta u} \int_0^t e^{-(h_u(t)-h_u(s))} ds, \quad h_u(t) := \int_0^t \frac{d}{K + c^\top x_u(\tau)} d\tau. \quad (42)$$

Since $\dot{p}_u(0) = b$, we have $c^\top p_u(s) \sim (c^\top b)s$, as $s \rightarrow 0$, and $e^{As} \sim I$, as $s \rightarrow 0$. Therefore, for each fixed $t > 0$,

$$\begin{aligned} h_u(t) &:= \int_0^t \frac{d}{K + c^\top x_u(\tau)} d\tau = \int_0^t \frac{d}{K + c^\top e^{A\tau}x_0 + uc^\top p_u(\tau)} d\tau \\ &= \int_0^t \frac{d}{K + c^\top x_0 + u(c^\top b)\tau} d\tau \\ &= \frac{d}{u(c^\top b)} \log \left(\frac{K + c^\top x_0 + u(c^\top b)t}{K + c^\top x_0} \right) \sim \frac{d}{u(c^\top b)} \log u + O(u^{-1}) \end{aligned}$$

as $u \rightarrow \infty$. Since $h_u \rightarrow 0$, using Taylor expansion $e^{-z} = 1 - z + O(z^2)$ as $z \rightarrow 0$, we have $e^{-h_u(t)} = 1 - h_u(t) + O(h_u(t)^2)$. Using (42), it follows that

$$y_u(t) = y_0 - \frac{dy_0}{u(c^\top b)} \log u + O(u^{-1}), \quad u \rightarrow \infty. \quad (43)$$

From (43), we can solve using two alternative ways:

- Consider (43), it follows that

$$\frac{y_0 - y_u(t)}{u} = \frac{dy_0}{u^2(c^\top b)} \log u + O(u^{-2}). \quad (44)$$

On the other hand, by (41),

$$\frac{d}{u} \int_0^t \frac{(K + c^\top e^{As}x_0)y_u(s)}{(K + c^\top x_u(s))^2} ds = O(u^{-2}), \quad (45)$$

because y_u remains bounded. Therefore, by (41), (44), and (45)

$$G_u(t) = \frac{dy_0}{u^2(c^\top b)} \log u + O(u^{-2}), \quad u \rightarrow \infty. \quad (46)$$

In particular, for every fixed $t > 0$, one has $G_u(t) > 0$ for all sufficiently large u .

- Differentiating (43) with respect to u , we obtain

$$q_u(t) = \frac{dy_0}{c^\top b} \frac{\log u}{u^2} + O(u^{-2}), \quad u \rightarrow \infty.$$

In particular, for every fixed $t > 0$, one has $q_u(t) > 0$ for all sufficiently large u . Hence $\partial_u \text{cDR}(u, T) > 0$ for all sufficiently large u .

Thus there exist $u_-, u_+ > 0$ such that $\partial_u \text{cDR}(u_-, T) < 0$ and $\partial_u \text{cDR}(u_+, T) > 0$, and therefore $\text{cDR}(u, T)$ is not monotone.

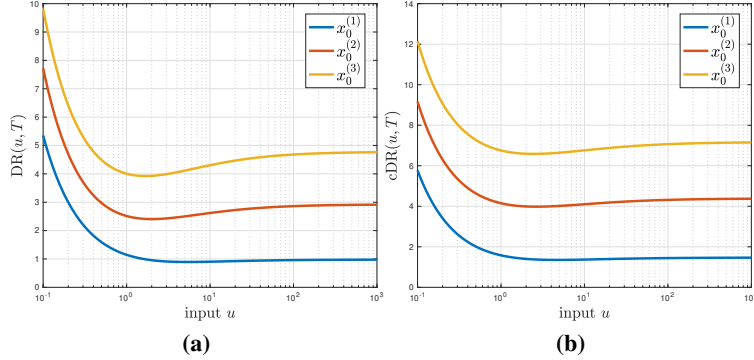


Figure 15: Vector System 7: Dose response $\text{DR}(u, T)$ and cumulative dose response $\text{cDR}(u, T)$ for $u \in [10^{-1}, 10^3]$. Results are shown for three fixed initial conditions x_0 . The curves illustrate that $\text{cDR}(u, T)$ is not monotone with respect to u , exhibiting a change in monotonicity.

Vector case of System 8

We consider the vector analogue of System 8:

$$\dot{x}_u = Ax_u + bu, \quad \dot{y}_u = d - \frac{\beta u}{K + c^\top x_u} y_u, \quad (47a)$$

where $A \in \mathbb{R}^{n \times n}$ is Metzler and Hurwitz, $b, c \in \mathbb{R}_+^n$, and $K, \beta, d \in \mathbb{R}_+$. We assume that the initial condition $x_u(0) = x_0$ is independent of u , with $x_0 \geq 0$, and that $y_u(0) = y_0$ is also independent of u .

Let $q_u := \partial_u y_u \in \mathbb{R}$. Differentiating the y -equation in (47a) with respect to u gives

$$\dot{q}_u = -\frac{\beta}{K + c^\top x_u} y_u + \frac{\beta u c^\top p_u}{(K + c^\top x_u)^2} y_u - \frac{\beta u}{K + c^\top x_u} q_u, \quad q_u(0) = 0.$$

Therefore, q_u satisfies $\dot{q}_u + a_u q_u = g_u$, $q_u(0) = 0$ with

$$a_u = \frac{\beta u}{K + c^\top x_u}, \quad g_u = -\frac{\beta}{K + c^\top x_u} y_u + \frac{\beta u c^\top p_u}{(K + c^\top x_u)^2} y_u.$$

Using (34), we rewrite g_u as

$$g_u = \frac{\beta y_u}{(K + c^\top x_u)^2} (u c^\top p_u - K - c^\top x_u) = -\frac{\beta y_u}{(K + c^\top x_u)^2} (K + c^\top e^{At} x_0). \quad (47b)$$

Since $K > 0$, $e^{At} \geq 0$, $c \in \mathbb{R}_+^n$, $x_0 \geq 0$, and $y_u \geq 0$, we conclude that $g_u \leq 0$, $\forall t \geq 0$. Integrating g_u over $[0, t]$, we obtain

$$G_u := \int_0^t g_u(s) ds = -\int_0^t \frac{\beta y_u(s)}{(K + c^\top x_u(s))^2} (K + c^\top e^{As} x_0) ds \leq 0. \quad (47c)$$

Since $a_u = \beta u / (K + c^\top x_u) \geq 0$, the kernel λ_u defined in Theorem 1 satisfies

$$\lambda_u = \int_t^T \exp\left(-\int_t^s \frac{\beta u}{K + c^\top x_u(\tau)} d\tau\right) ds. \quad (47d)$$

Therefore, $\lambda_u \geq 0$, $\forall t \in [0, T]$.

Proposition 15 Consider the system (47a), where A is Metzler and Hurwitz, $b, c \in \mathbb{R}_+^n$, $K, \beta, d \in \mathbb{R}_+$, and $x_0 \geq 0$ is independent of u . Then, for every $T > 0$, the $\text{cDR}(u, T)$ is monotone nonincreasing with respect to u .

Proof 16 The sensitivity $q_u(t) = \partial_u y_u(t)$ satisfies $\dot{q}_u(t) + \frac{\beta u}{K + c^\top x_u(t)} q_u(t) = g_u(t)$ in which $g_u(t)$ is given in (47b). Also, $\lambda_u(t) \geq 0, \forall t \in [0, T]$ by (47d). Thus $\lambda_u(t) g_u(t) \leq 0, \forall t \in [0, T]$. By Theorem 1 (i), $\partial_u \text{cDR}(u, T) \leq 0$. Hence $u \mapsto \text{cDR}(u, T)$ is monotone nonincreasing.

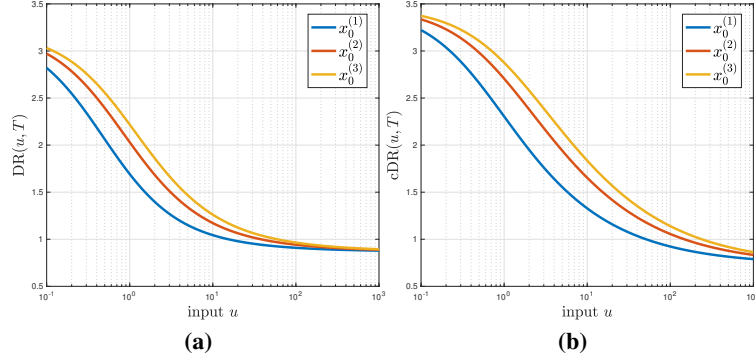


Figure 16: Vector System 8: Dose response $\text{DR}(u, T)$ and cumulative dose response $\text{cDR}(u, T)$ for $u \in [10^{-1}, 10^3]$. Results are shown for three fixed initial conditions x_0 . The curves illustrate that $\text{cDR}(u, T)$ is monotone nonincreasing with respect to u .